Foundations of Large-Scale Multimedia Information Management and Retrieval

Lecture #2 Feature Extraction

Edward Y. Chang

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Foundations of Large-Scale Multimedia Information Management and Retrieval
Mathematics of Perception

Foundations of Large-Scale Multimedia Information Management and Retrieval *Mathematics of Perception* covers knowledge representation and semantic analysis of multimedia data and scalability in signal extraction, data mining, and indexing. The book is divided into two parts: Part I - Knowledge Representation and Semantic Analysis focuses on the key components of mathematics of perception as it applies to data management and retrieval. These include feature selection/reduction, knowledge representation, semantic analysis, distance function formulation for measuring similarity, and multimodal fusion. Part II - Scalability Issues presents indexing and distributed methods for scaling up these components for high-dimensional data and Web-scale datasets. The book presents some real-world applications and remarks on future research and development directions.

The book is designed for researchers, graduate students, and practitioners in the fields of Computer Vision, Machine Learning, Large-scale Data Mining, Database, and Multimedia Information Retrieval.

Dr. Edward Y. Chang was a professor at the Department of Electrical & Computer Engineering, University of California at Santa Barbara, before he joined Google as a research director in 2006. Dr. Edward Y. Chang received his M.S. degree in Computer Science and Ph.D degree in Electrical Engineering, both from Stanford University.

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Mathematics of Perception

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Many Related Work on Feature Extraction

Knowledge or Feature extraction in Image processing: involves using algorithms to detect and isolate various desired portions or shapes (features) of a digitized image or video (for example to be used in Optical Character Recognition, Knowledge Based Road Extraction from Multisensor Imagery)

Low-level: Edge detection, Corner detection, Blob detection, Ridge detection, or more generally "Scale-invariant feature transform" (SIFT)

Curvature: Edge direction, changing intensity, shape information.

Image motion: Motion detection. Area based, differential approach. Optical flow.

Shape Based, Thresholding, Blob extraction

Template matching (medical imaging**)**

Hough transform: Lines, Circles/Ellipse, Arbitrary shapes (Generalized Hough Transform)

Flexible methods (for 2D, 3D or 3D+time edge extraction, road detection, MRI, fMRI)

- Deformable, parameterized shapes
- Active contours (snakes)

Features needed for high-level vision: recognition**,** classification, label objects

Four Representative Schemes

Focusing on "raw" features for now

- Color, Texture, Shape Features
- SIFT
- HMAX
- Convolutional Networks

- Later addressing semantics and context

Colors Textures Shapes

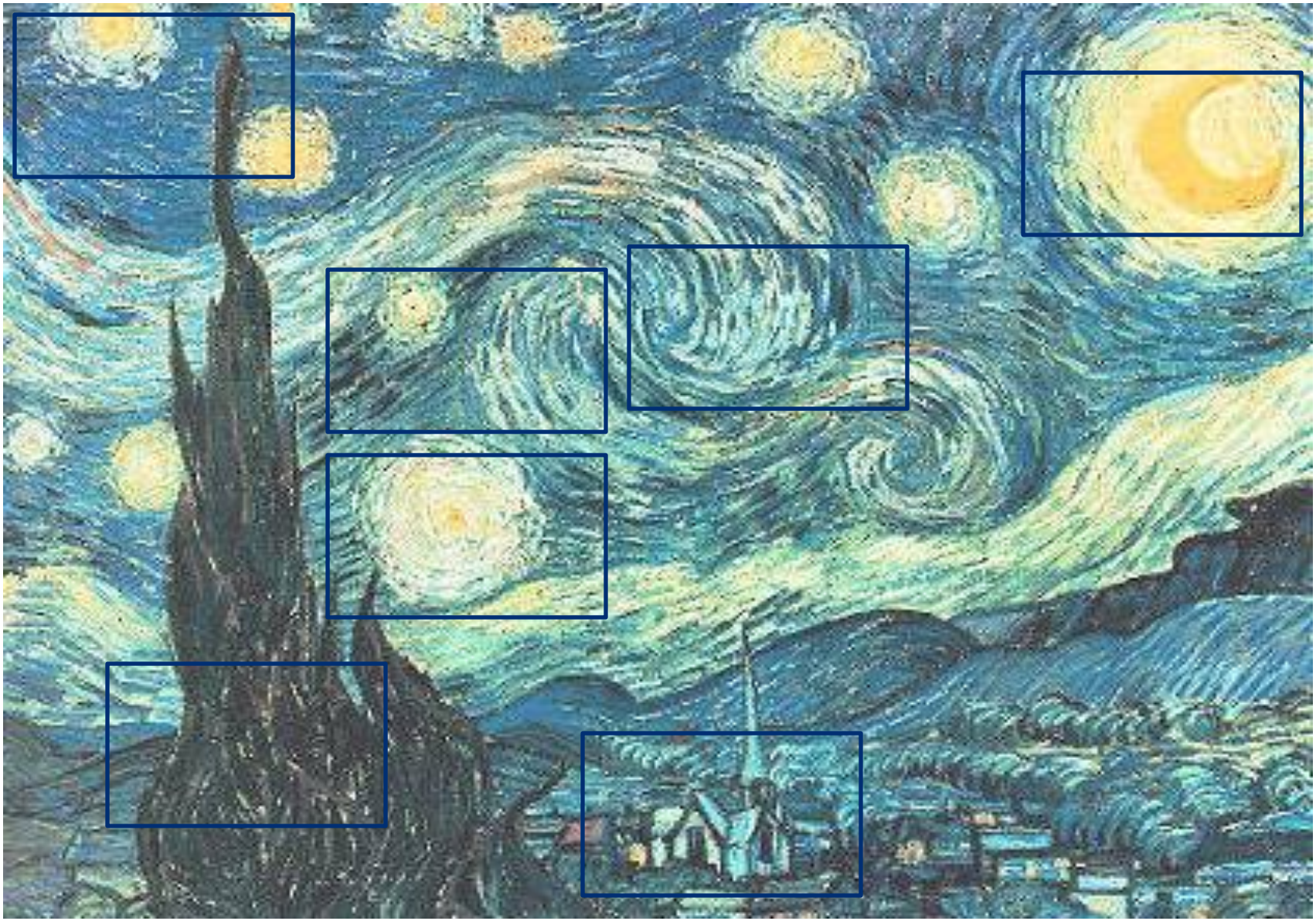
- Colors
  - Color histograms
  - Color correlation
- Shapes
  - Edges, Corner histograms
- Textures
  - Fourier descriptor
  - Multi-scale wavelets

Global Features…



Local Features?

- Divide an image into blocks
- Derive a code book (vocabulary) from blocks
- Employ LDA like algorithms



Key Issues

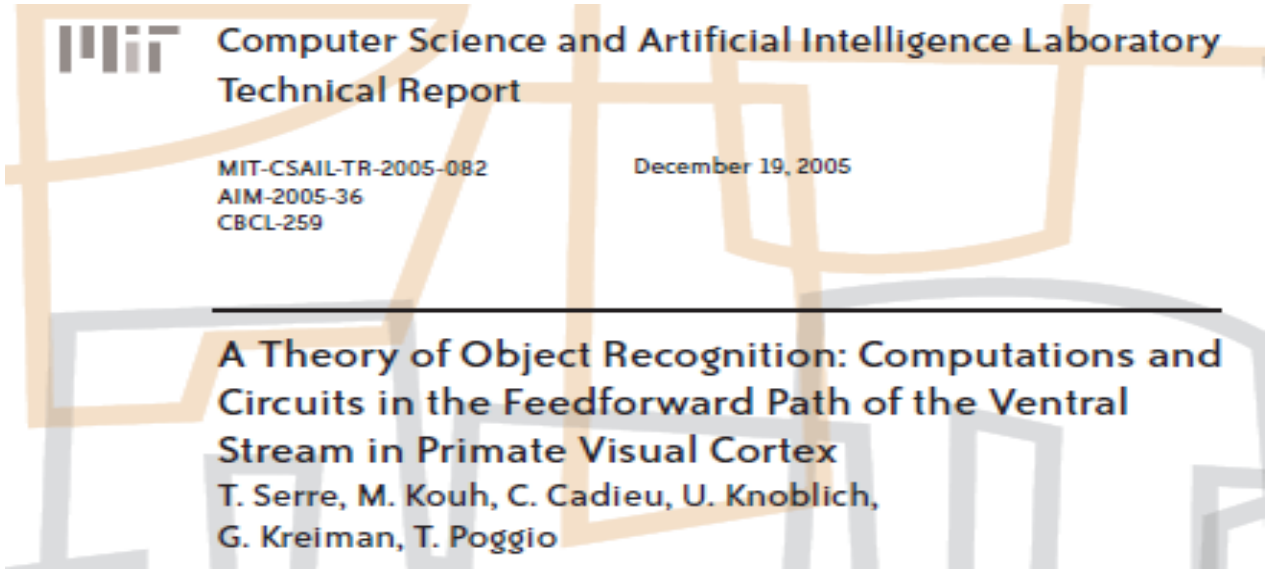
- Feature Invariance
  - Same object should have the same features
- Selectivity
  - Different objects should exhibit different features for telling them apart

- Challenges
  - Environmental factor (e.g., lighting condition, occlusion)
  - Equipment factor (e.g., different camera brands)
  - Algorithmic factor (e.g., how local is local?)
  - Aliasing


VAN GOGH GALLERY
www.vangoghgallery.com

Feature Invariance

- Human visual system works so well!
- "Mental" model (T. Serre, T. Poggio; MIT 2005)
  - Ventral visual pathway
  - Deep learning



Computer Science and Artificial Intelligence Laboratory
Technical Report

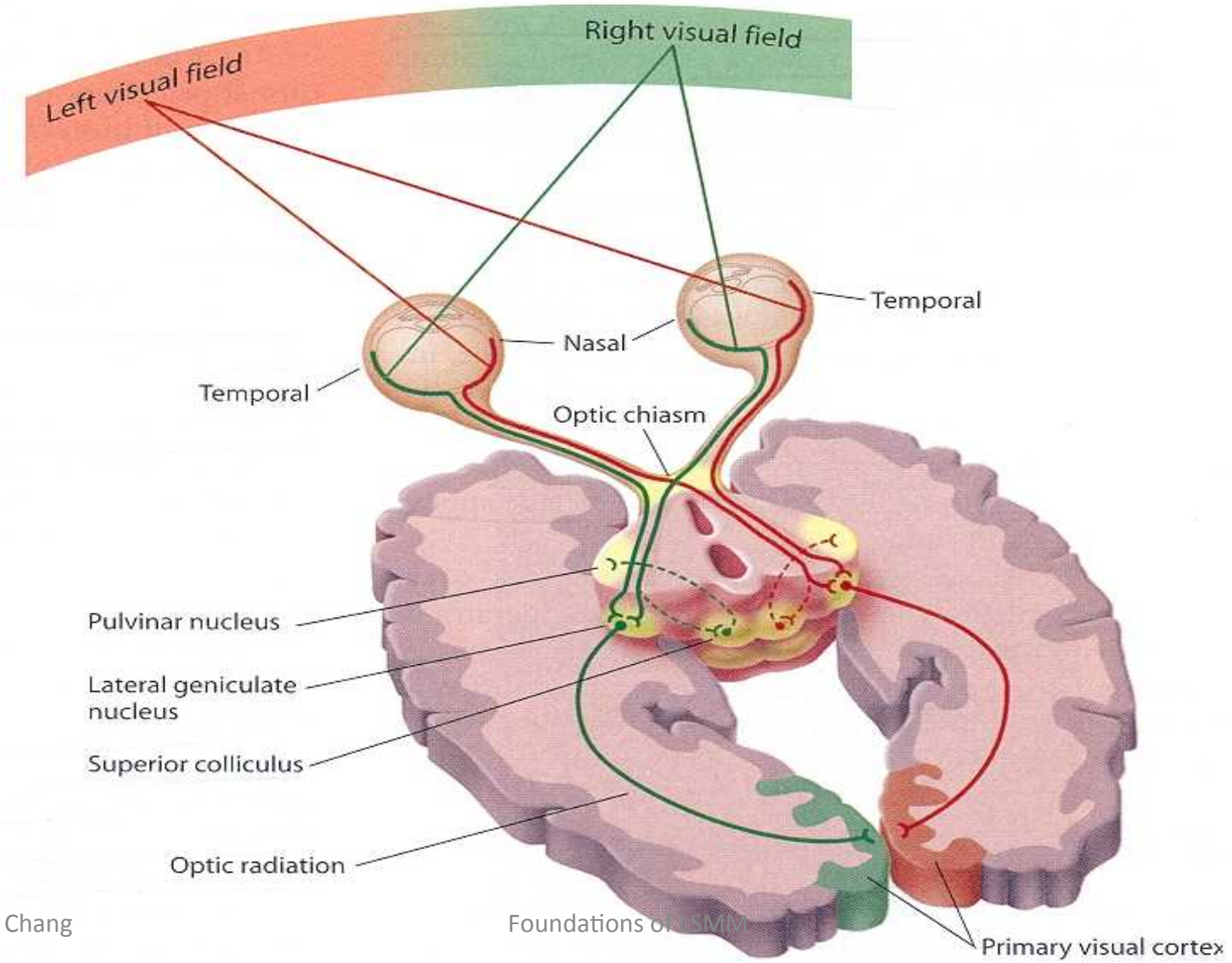
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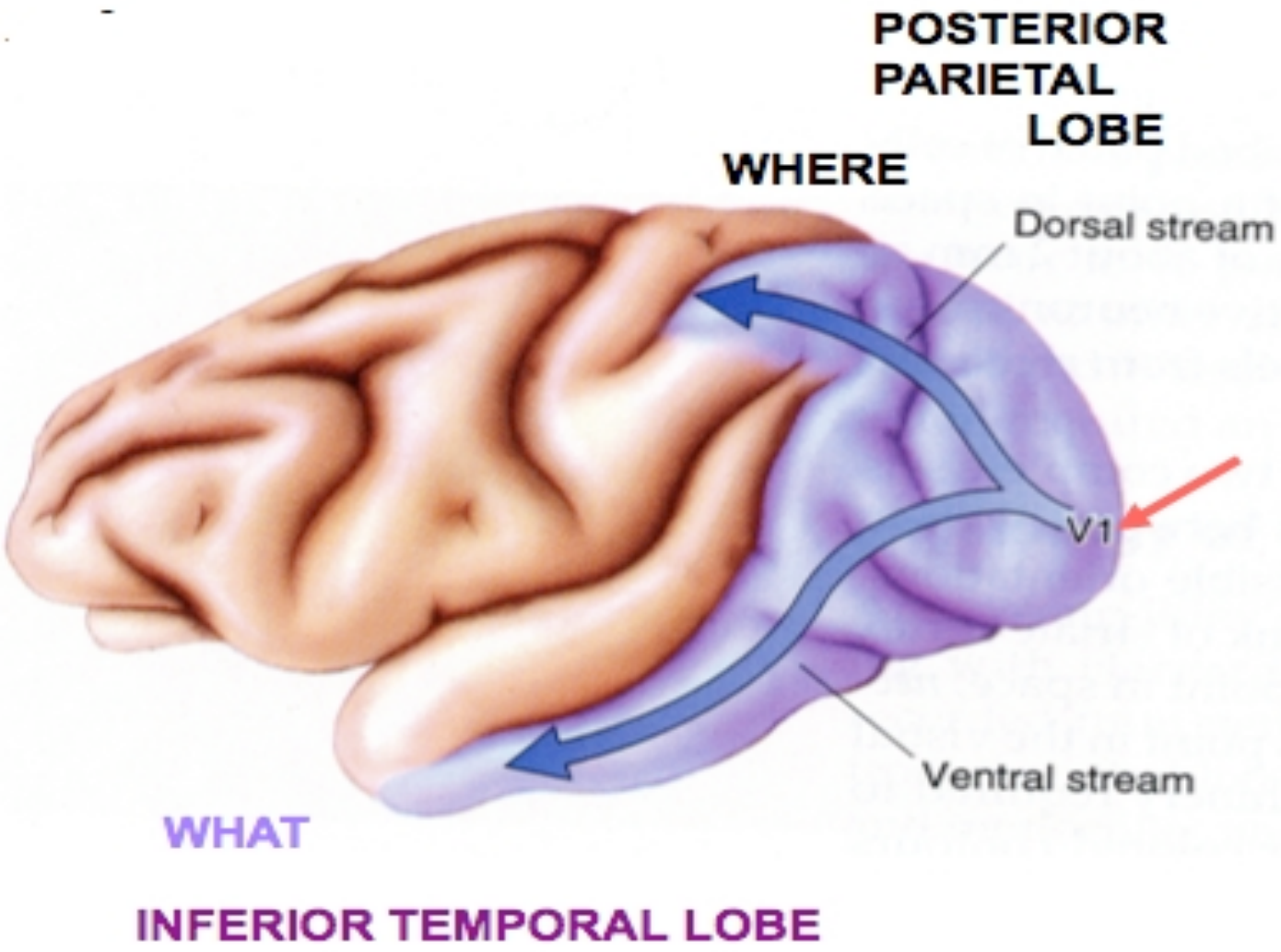
December 19, 2005

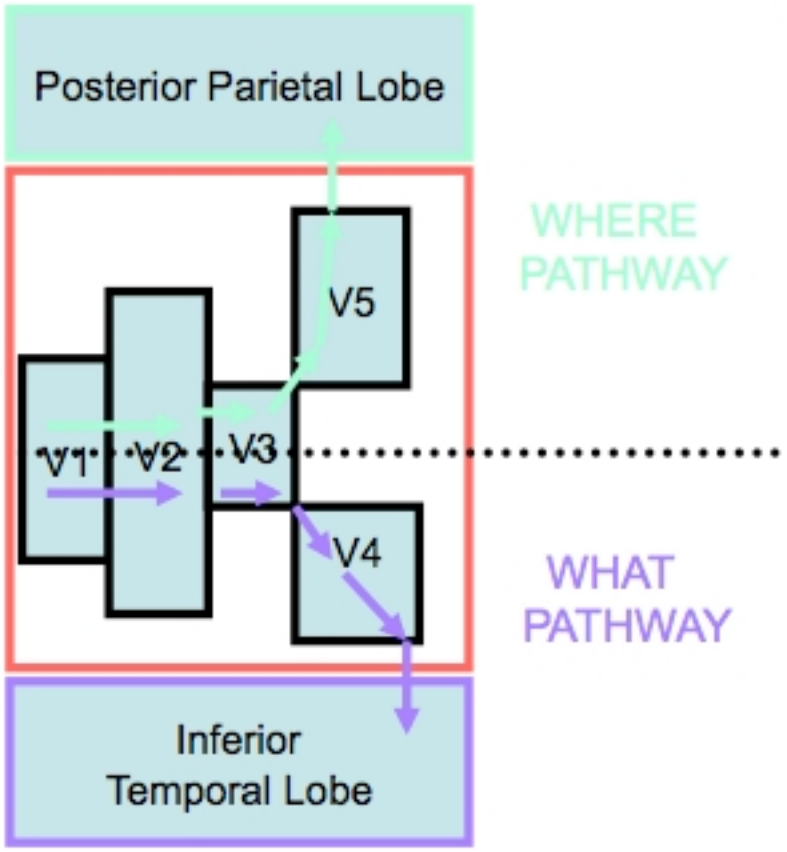
A Theory of Object Recognition: Computations and Circuits in the Feedforward Path of the Ventral Stream in Primate Visual Cortex

T. Serre, M. Kouh, C. Cadieu, U. Knoblich, G. Kreiman, T. Poggio

Visual Pathway [Hubel Wiesle, 68]




POSTERIOR
PARIETAL
LOBE
WHERE
Dorsal stream
V1
Ventral stream
WHAT
INFERIOR TEMPORAL LOBE



WHERE PATHWAY

WHAT PATHWAY

Areas of Occipital Cortex
VI – segregates pattern visior from motion signals

V2 – 3D vision/ seeing camouflage/ more complex patterns

V3 – shape perception

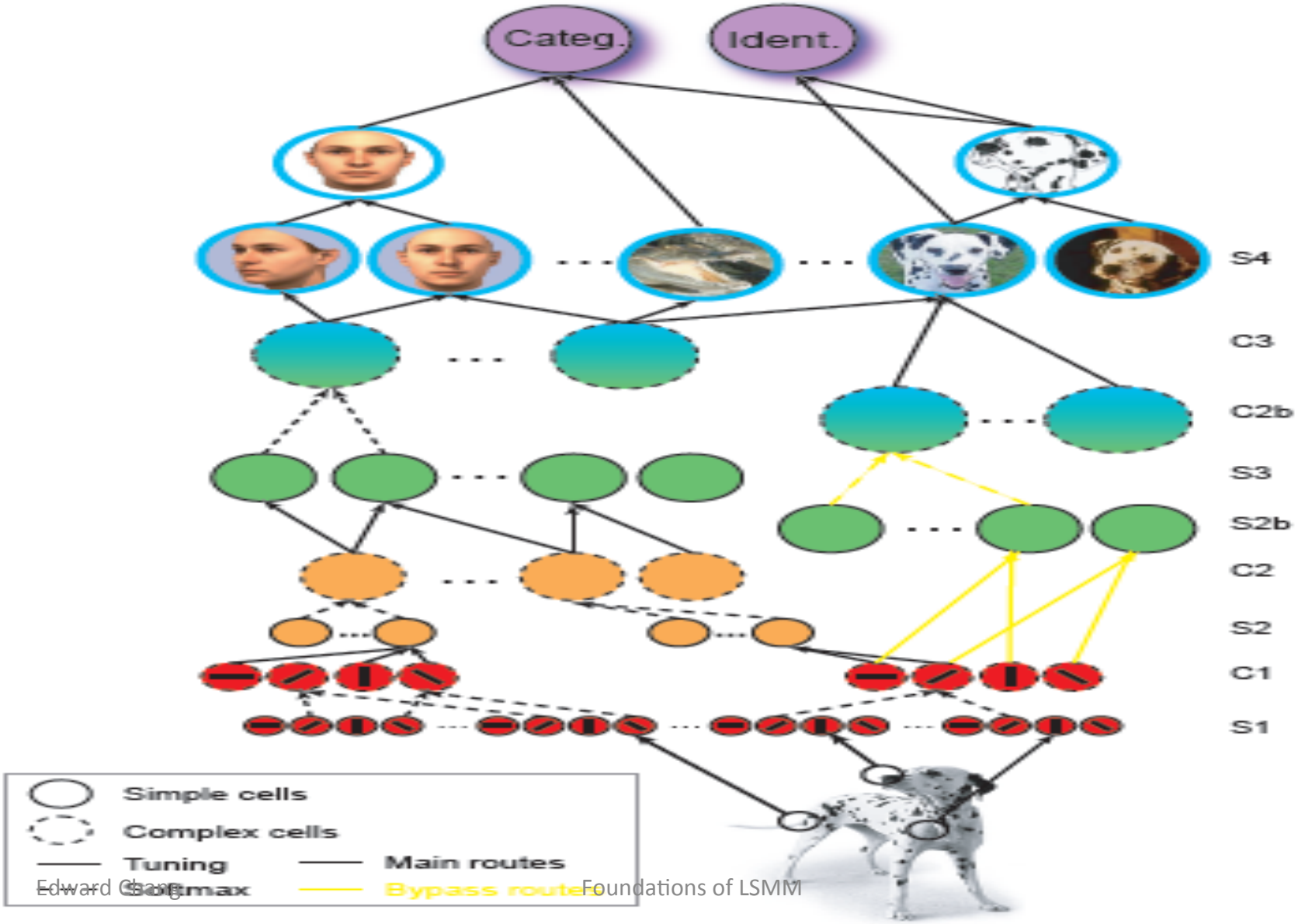
V4 – colour area and shape perception

V5 – motion area

Feedforward Path of Ventral Stream

- Invariance
  - Starting with scale/position invariance over a restricted range
  - Then invariance of view points and other transformations
- Multi-layer, multi-area
  - Improve complexity of optimal stimulus
- Feedforward
  - First 150 millisecond of perception
  - W/o feedback

Multi-layer Visual Pathway



Six Steps of HMAX [T. Serre, T. Poggio; MIT 2005]

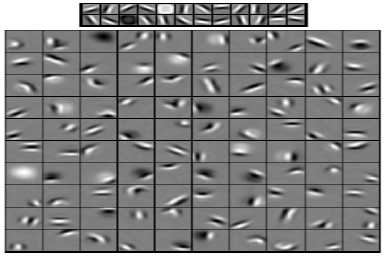
- Edge Detection, multi-/scale,direction (on/off, simple)
- Edge Pooling (max, invariance)
  - Keep "strong" features"
- Unsupervised Clustering (or)
  - Clustering edges into patches
- Part Detection (on/off, simple)
  - Find matching patches in photos
- Part Pooling (max, invariance)
  - Identify useful patches/parts
- Supervised Learning
  - Object ← parts

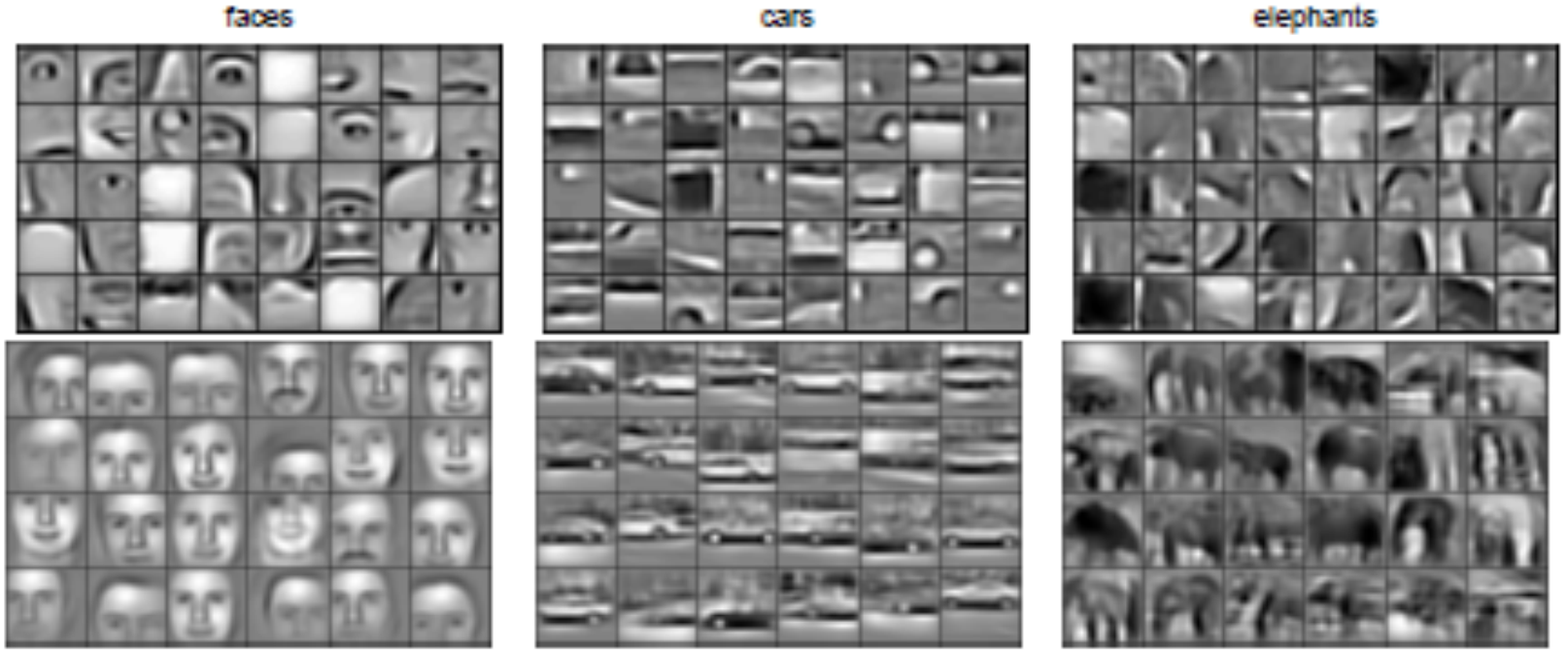
Limitations of Feedforward

- Forward only
- Missing attention
- Missing colors

Edges and Parts

Honglak Lee
Roger Grosse
Rajesh Ranganath
Andrew Y. Ng





Convolutional Networks [LeCun 97]

- Convolution (feature detection)
- Sub-sampling (multi-scale)
- Perform C & S iteratively to form a deep-learning network
- Learning weights from data

Five Layer Example

http://deeplearning.net/tutorial/lenet.html

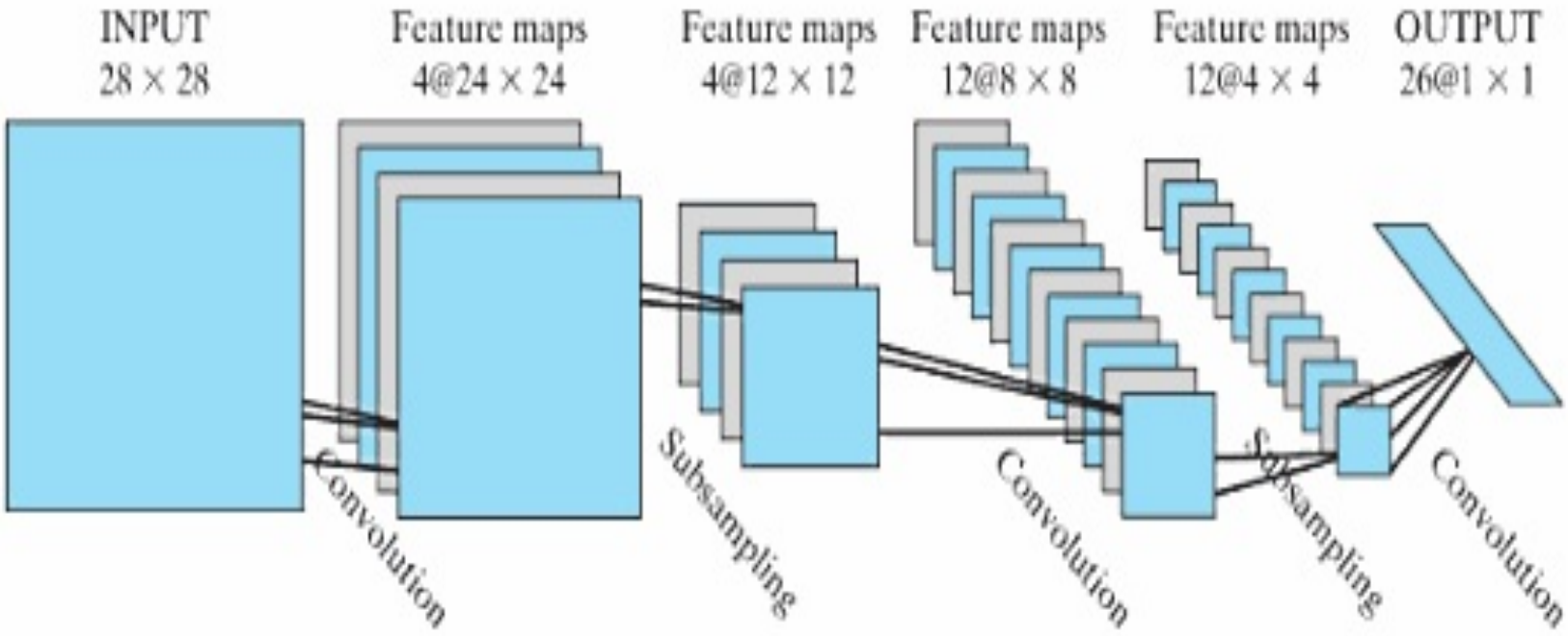
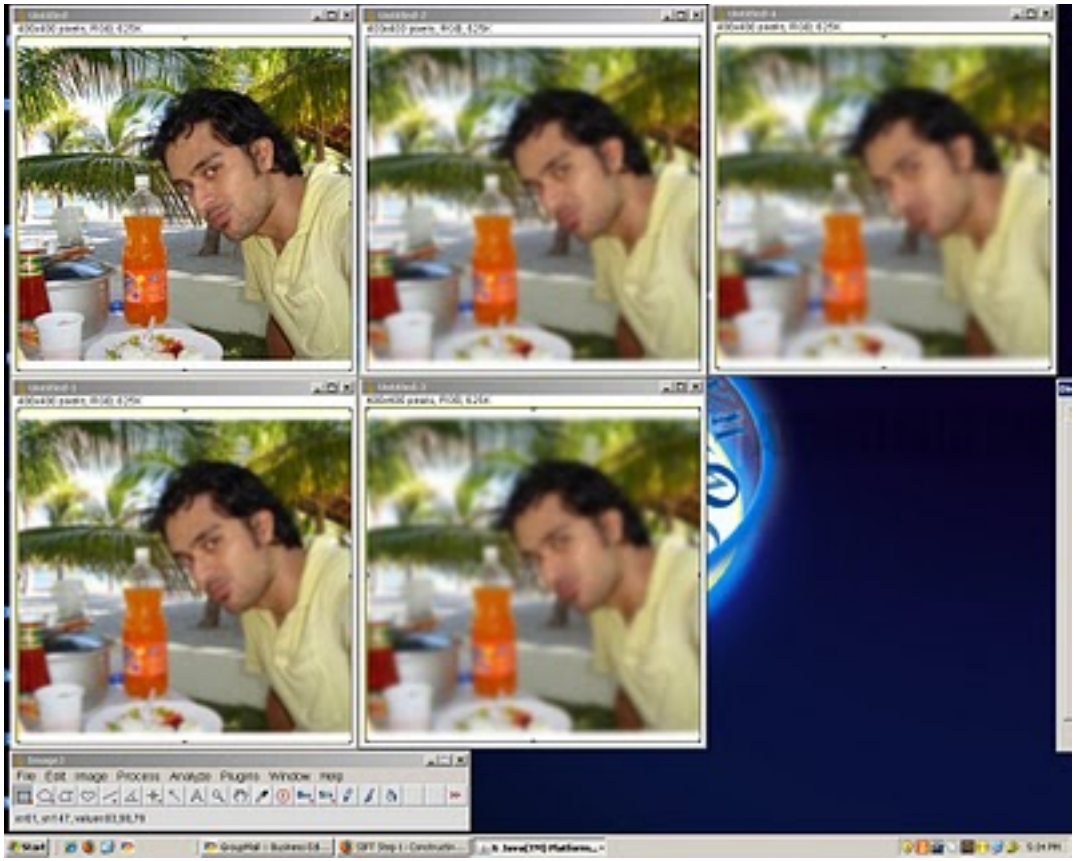


FIGURE 4.23 Convolutional network for image processing such as handwriting recognition. (Reproduced with permission of MIT Press.)

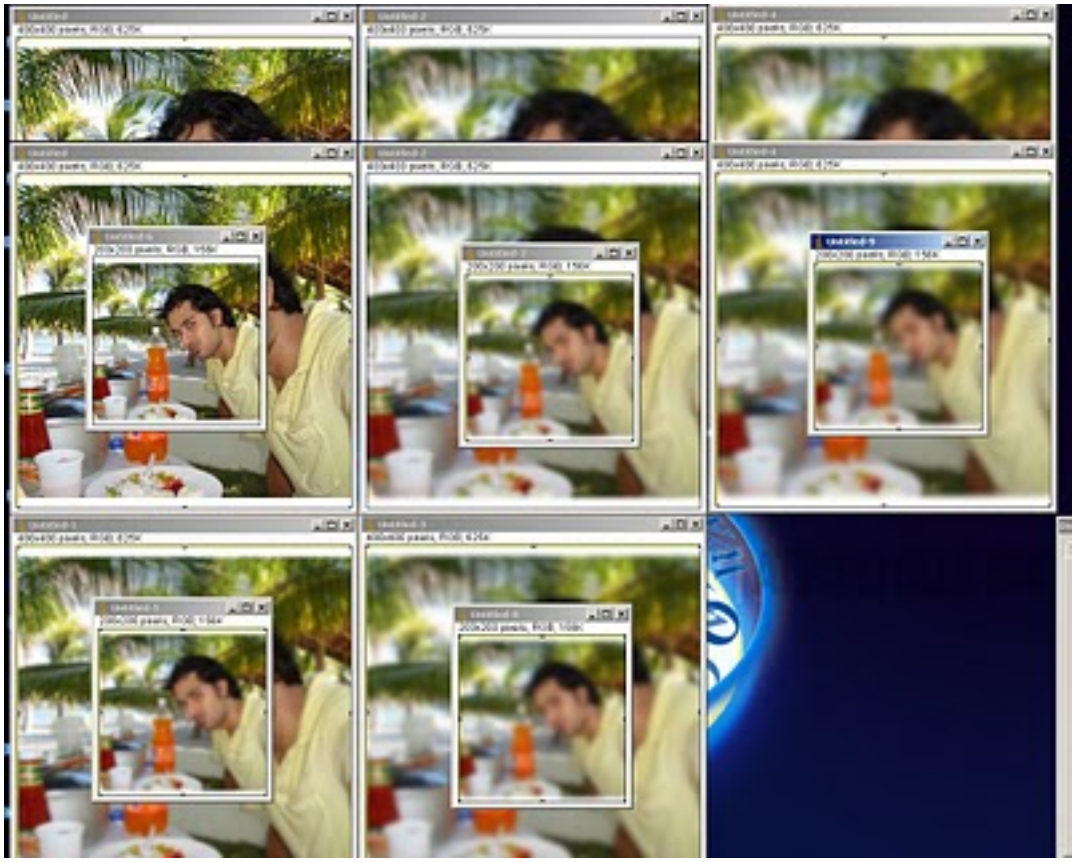
SIFT

- Scale-space Extrema Detection
  - Using DoG (Difference of Gaussian function)
  - Searching the image in all scales and locations for keypoints
  - DoG at (x,y) is $D(x; y;\sigma) = L(x; y; k\sigma) - L(x; y; \sigma)$
- Keypoint Localization
- Orientation Assignment
- Keypoint Descriptor

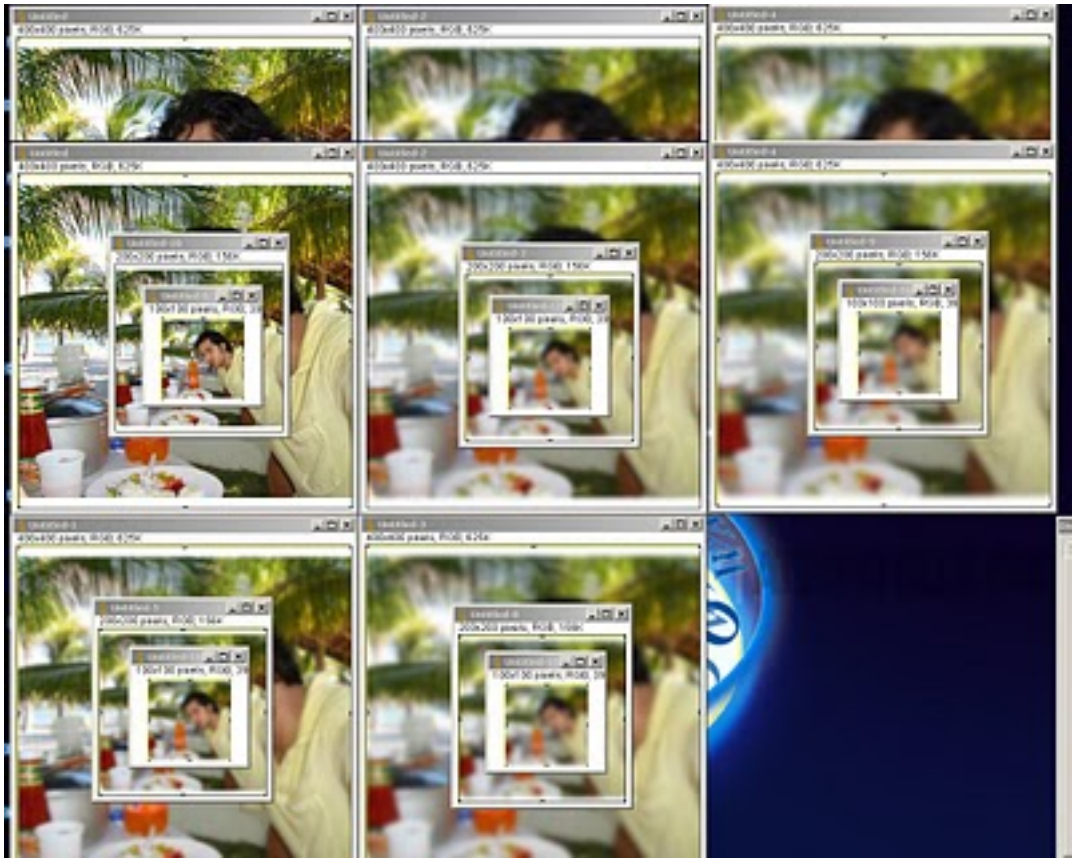
SIFT Step #1
Convolution, or Blurring



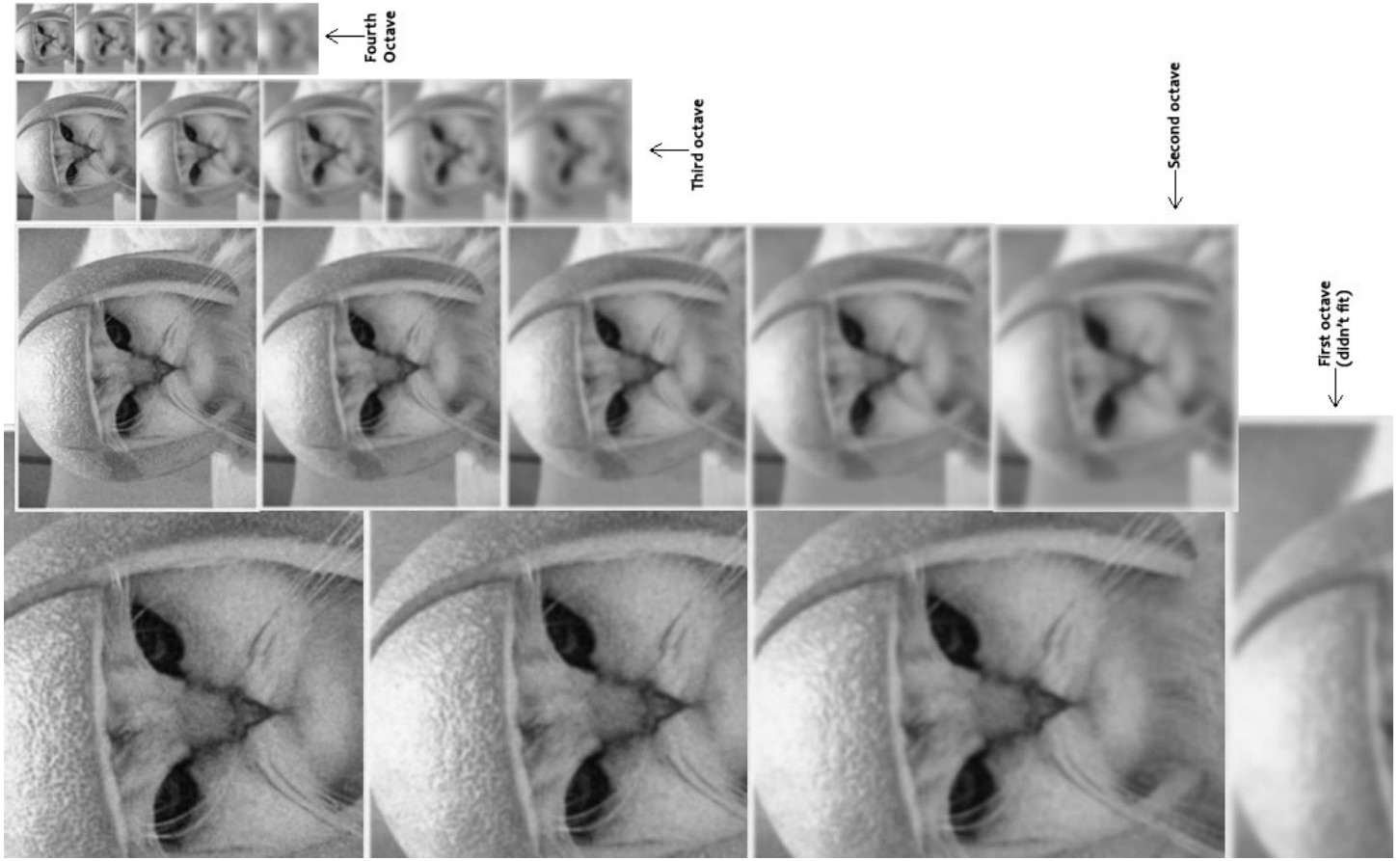
Multi-Scale



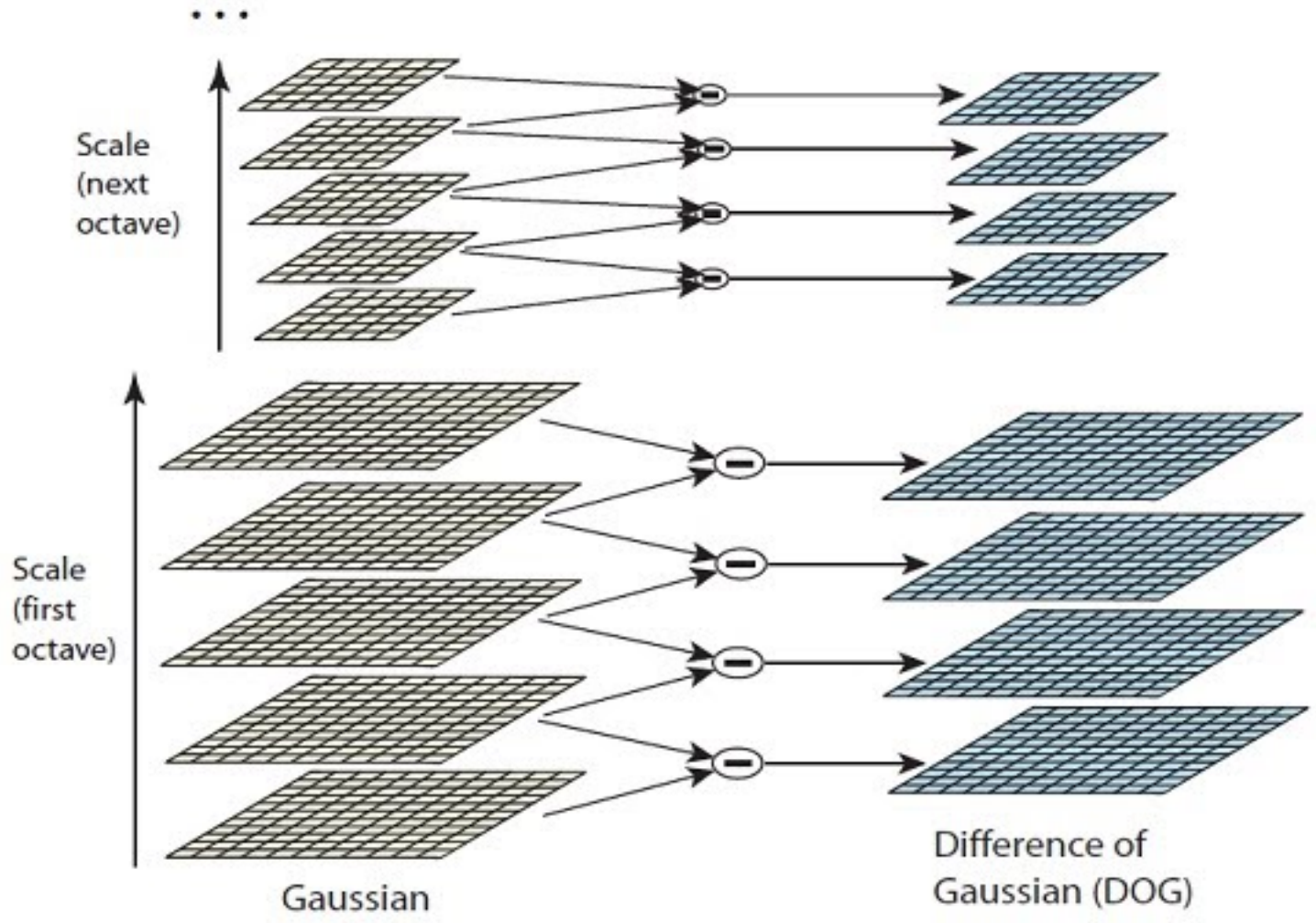
Multi-Scale



Another Example, 4x5 is good

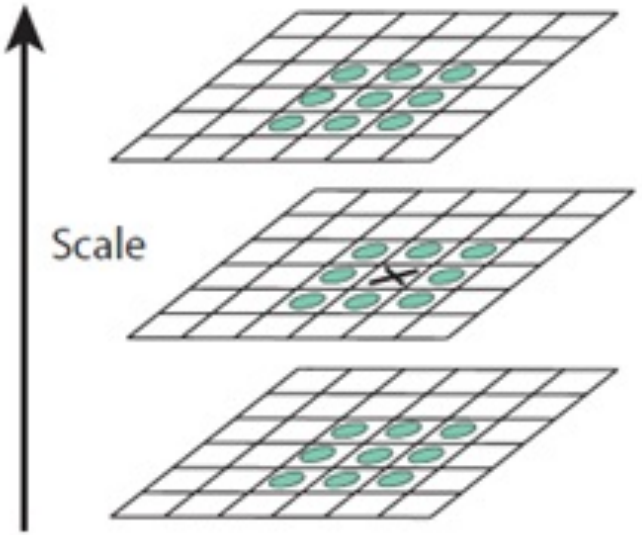


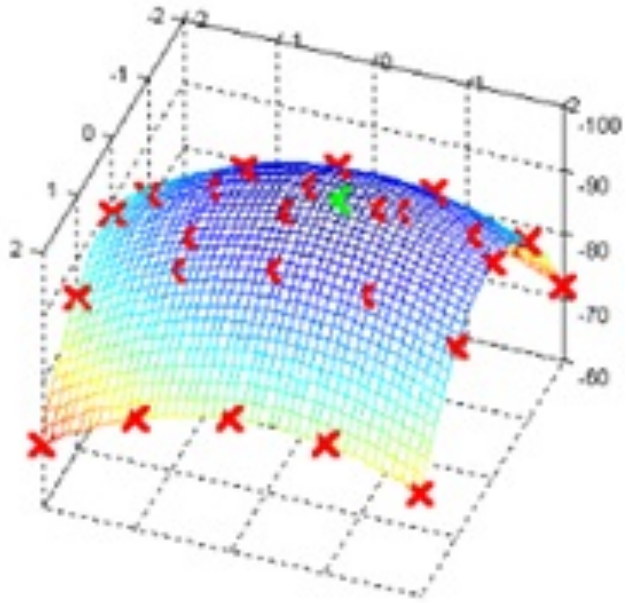
Scale Space Generated by DoG



Keypoints, pixels and subpixels

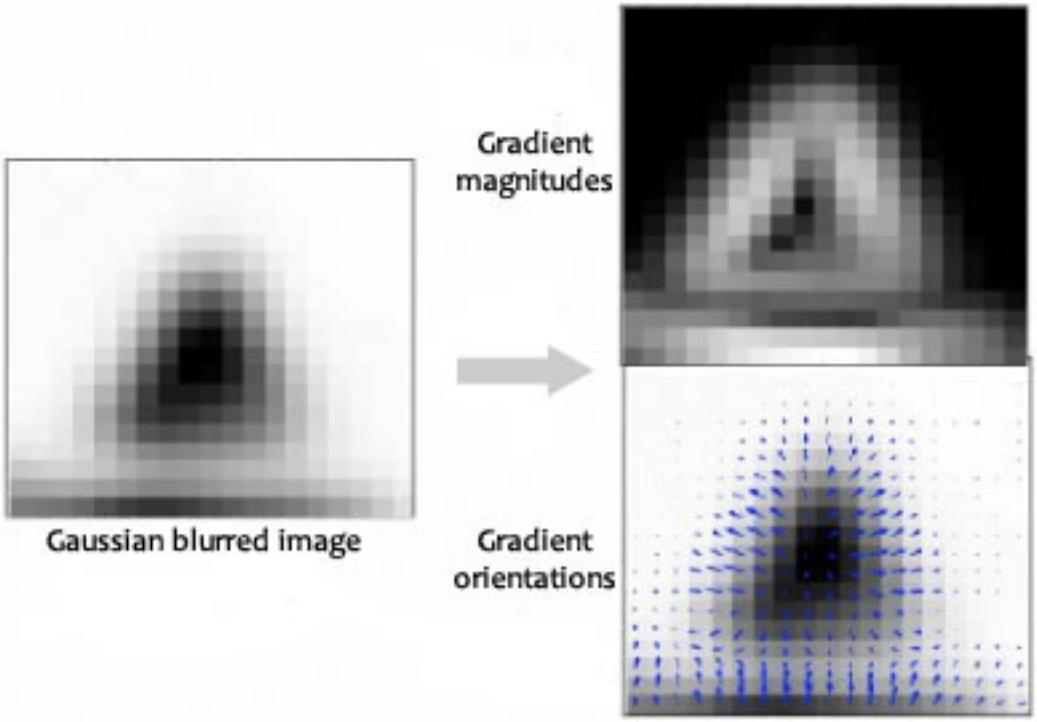
http://www.aishack.in/



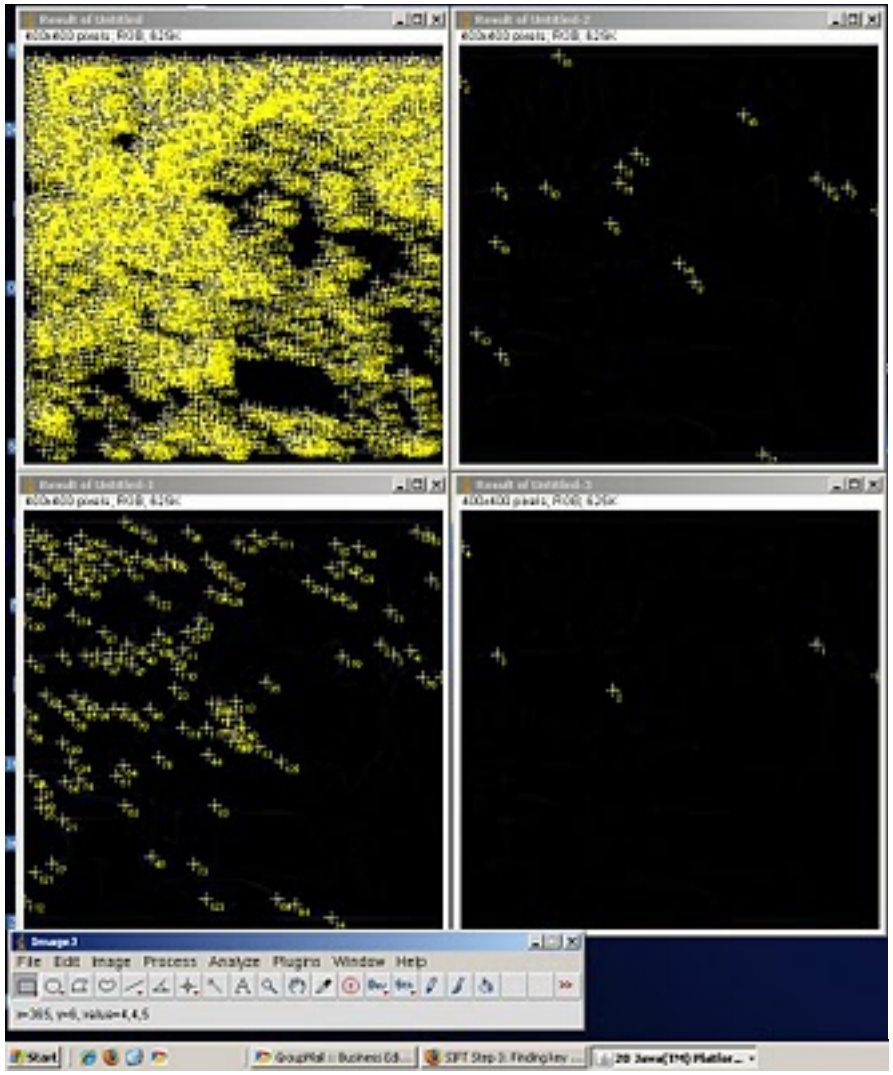


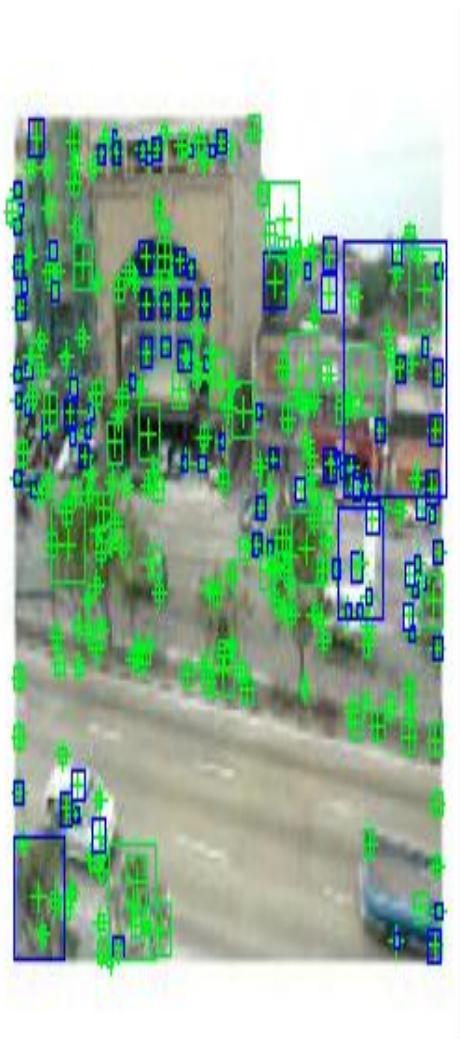
Reduce Keypoints and Assign Direction

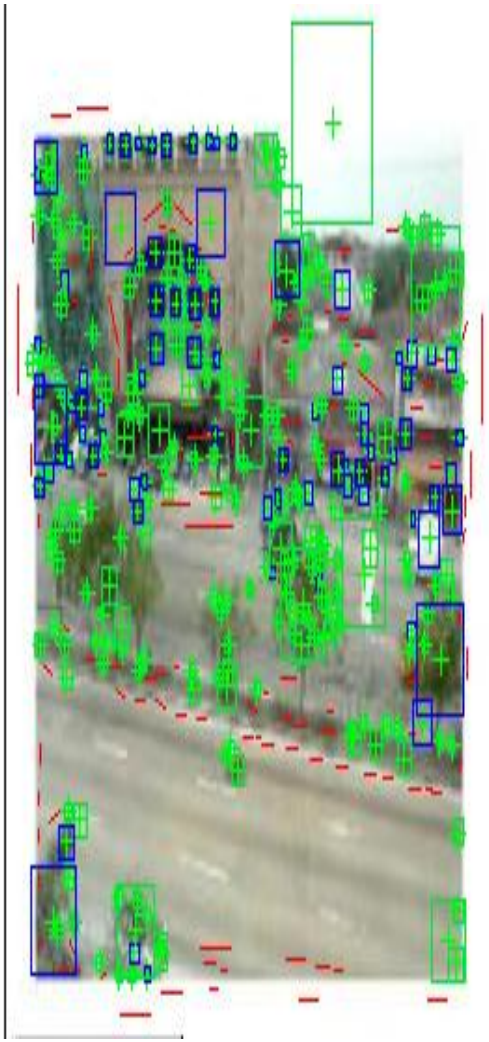
- Remove edges and low intensity points
- Compute gradients



SIFT Examples

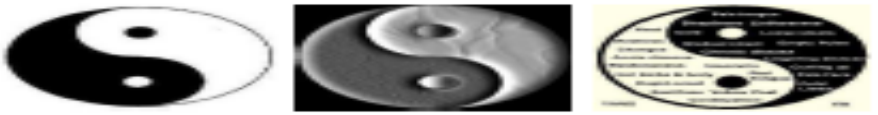
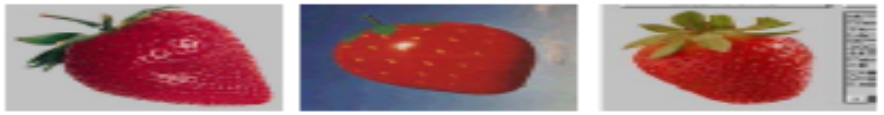




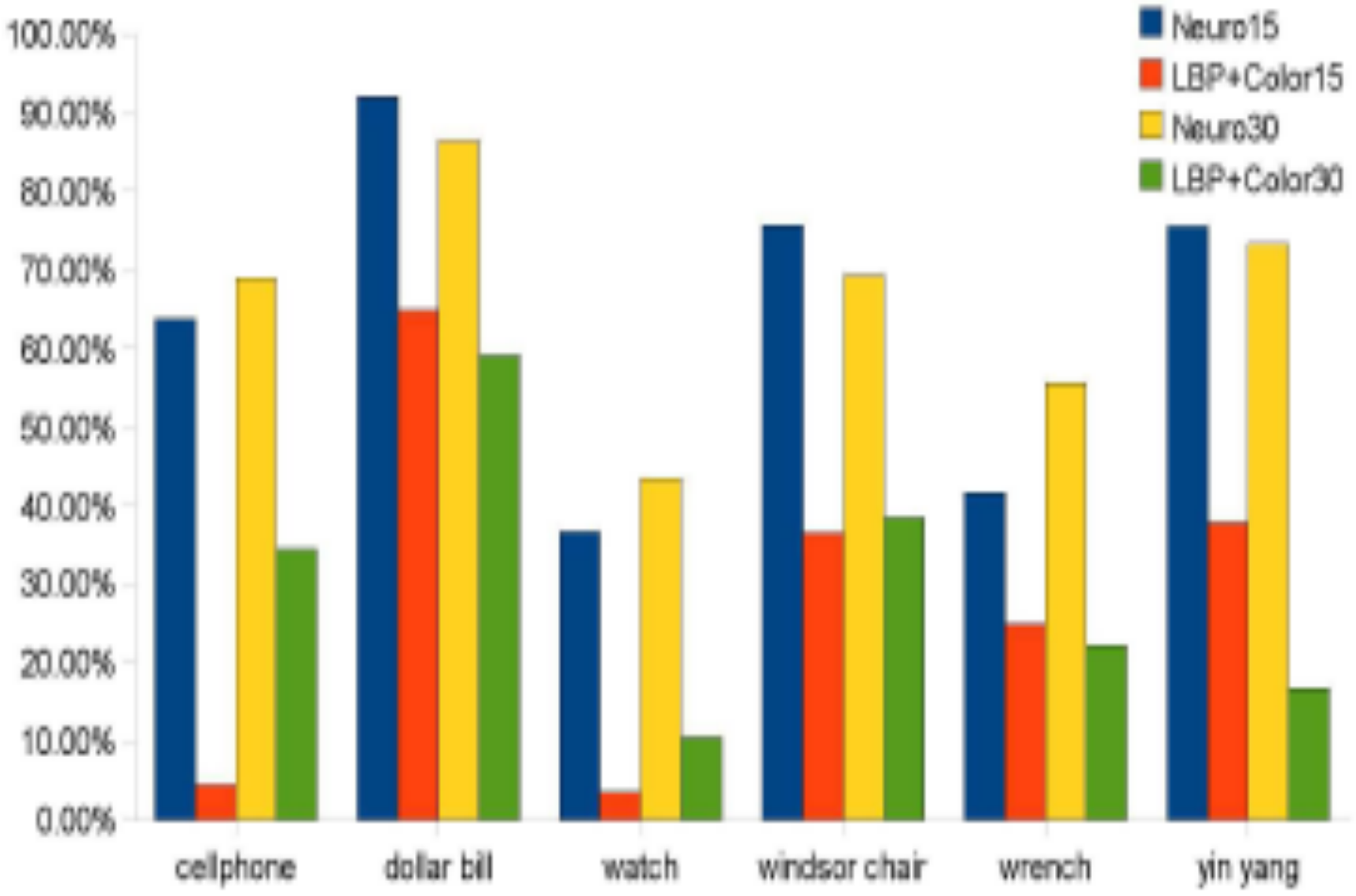


Results on Small Samples

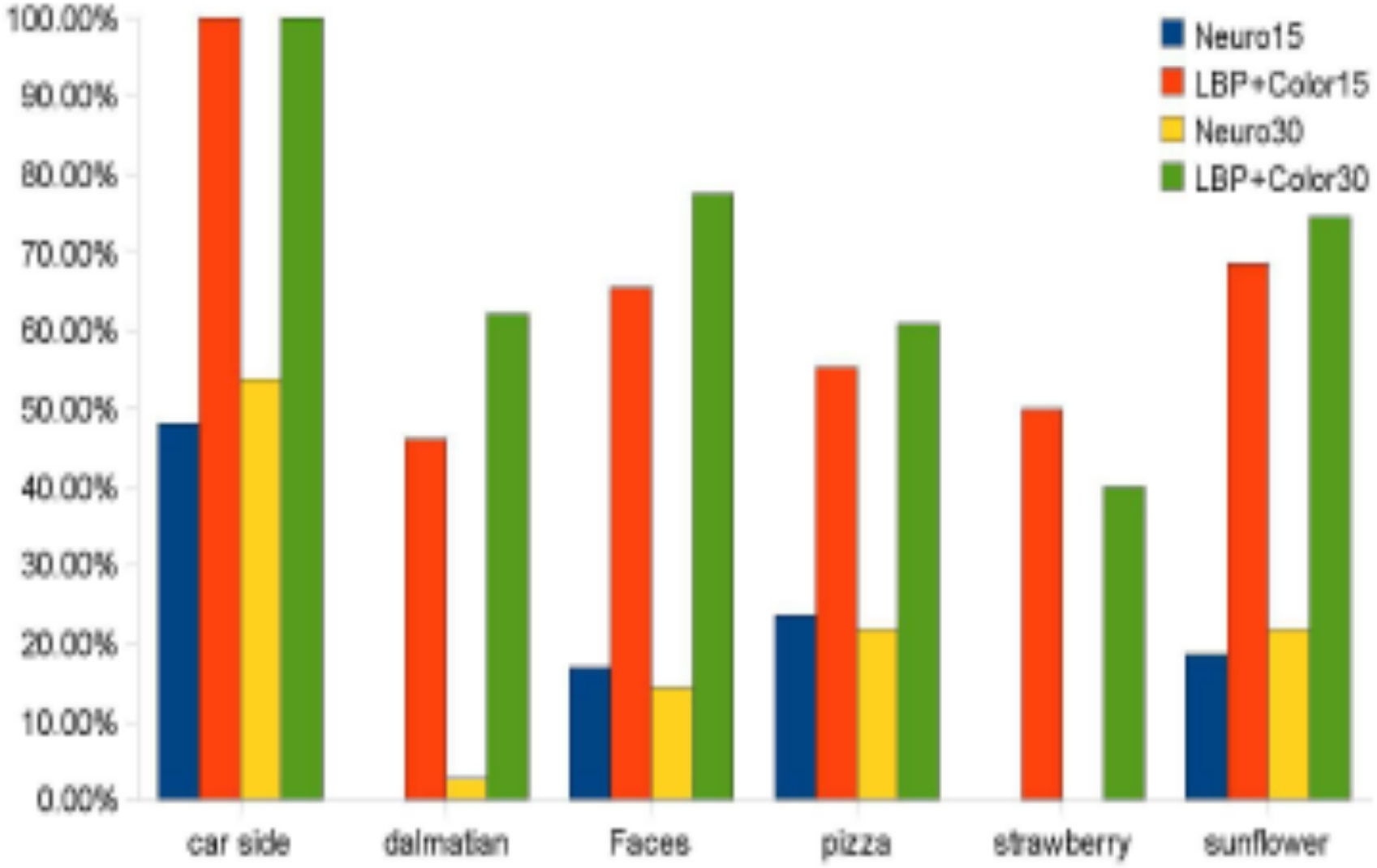
Table 4: Examples of Some Images

Class	Category	Images
global feature	cellphone	
	watch	
	yin_yang	
color feature	Faces	
	strawberry	
	sunflower	
texture feature	dalmatian	
	pizza	

Where Neural Model works better



Where LBP+color works better



More Results on ImageNet Dataset

A Data-Drive Study on Image Feature Extraction and Fusion, Z. Wang, F. Li, E. Y. Chang, and S. Yang, Google Technical Report, April 2012.

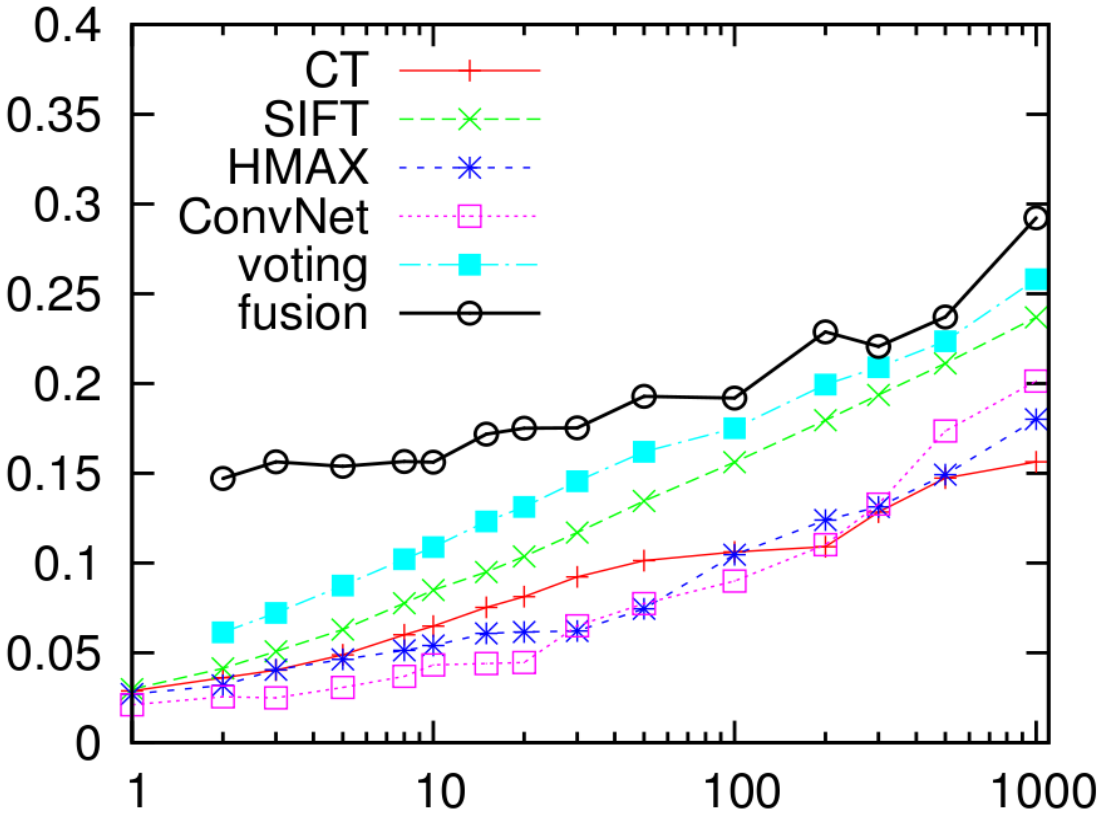
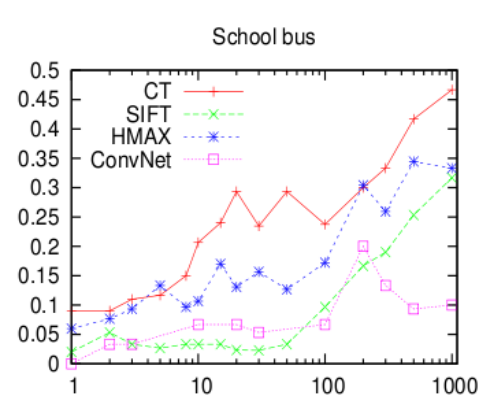
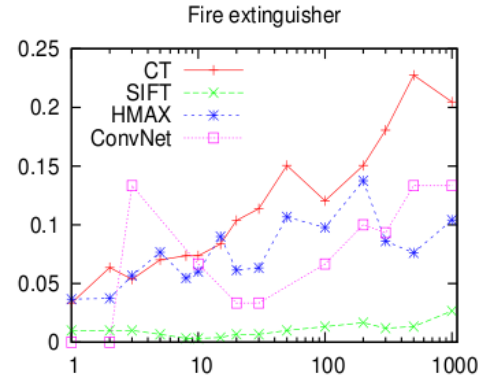
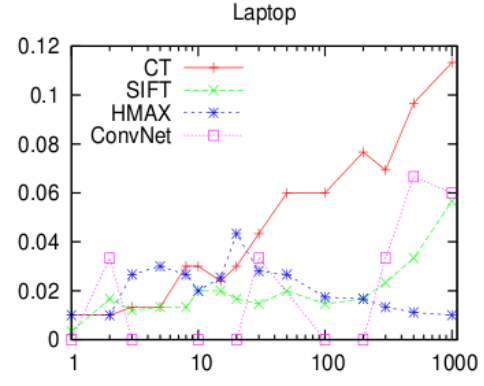


Fig. 7: Accuracy for ImageNet dataset. X-axis shows the number of training instances for each category. Y-axis shows the average accuracy.

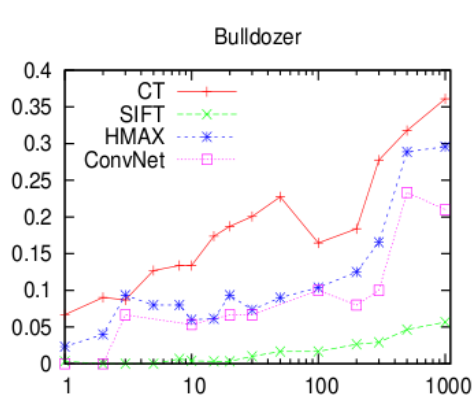
Color-Texture Codebook

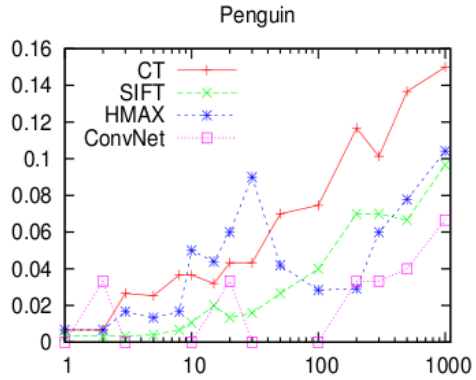


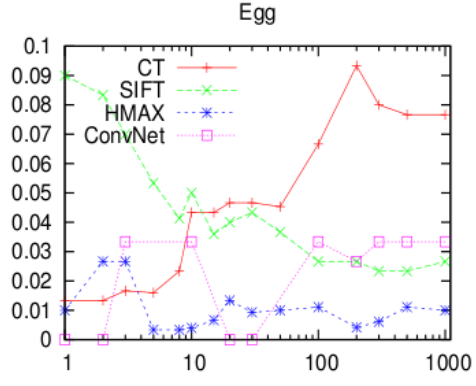






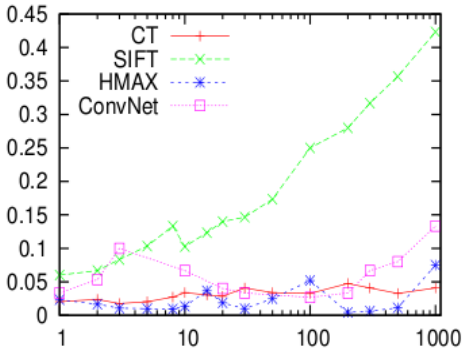




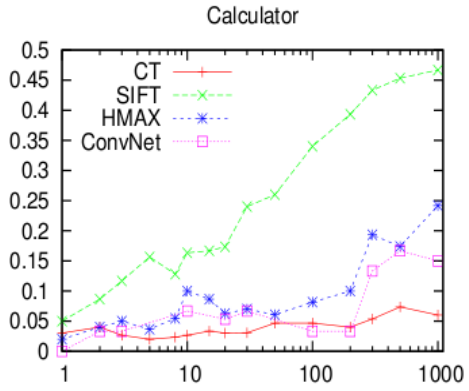




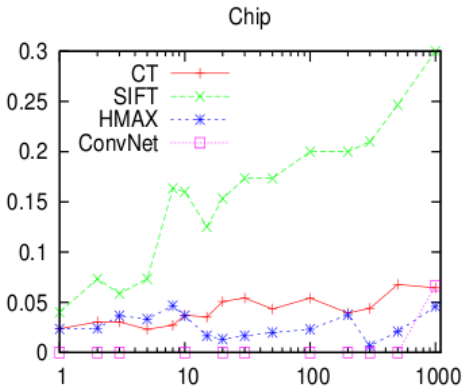
SIFT: matching by local shapes

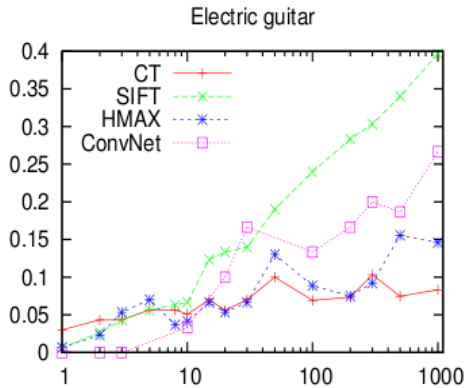


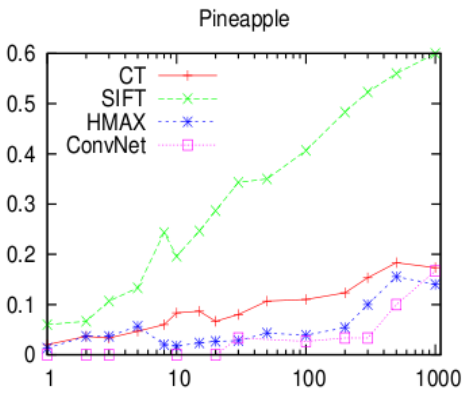


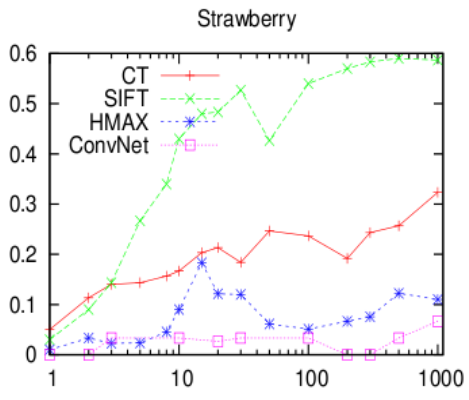




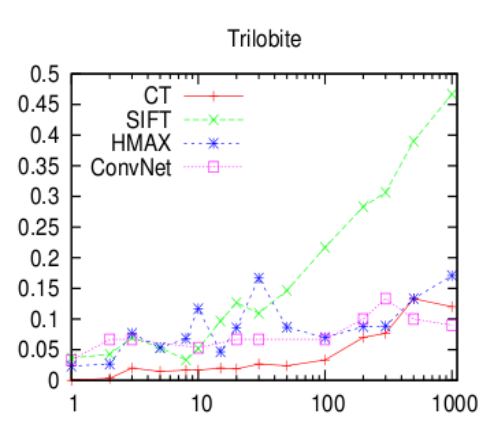


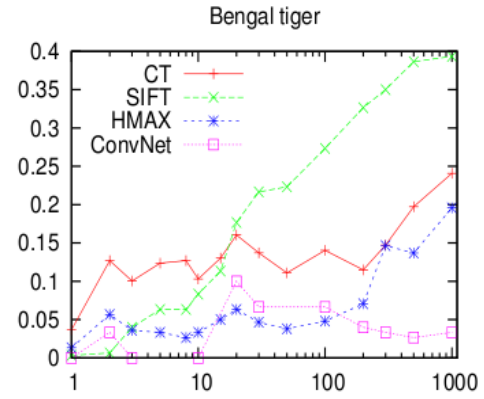


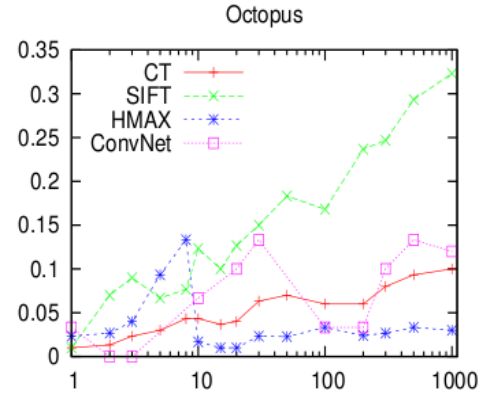




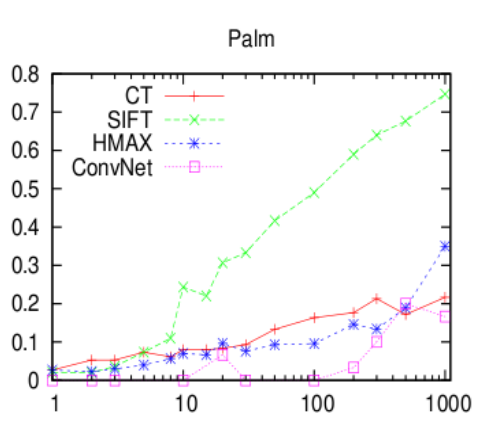
SIFT: matching by near-duplicate

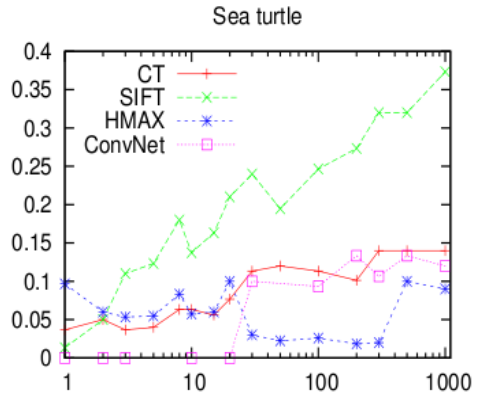


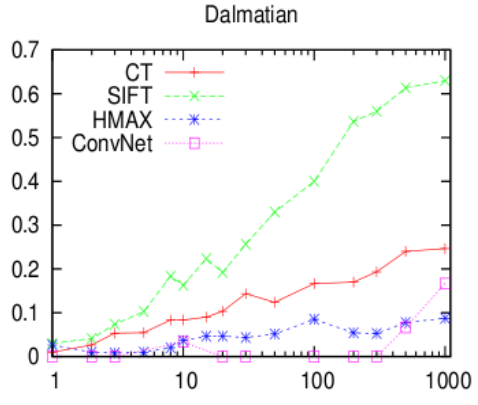






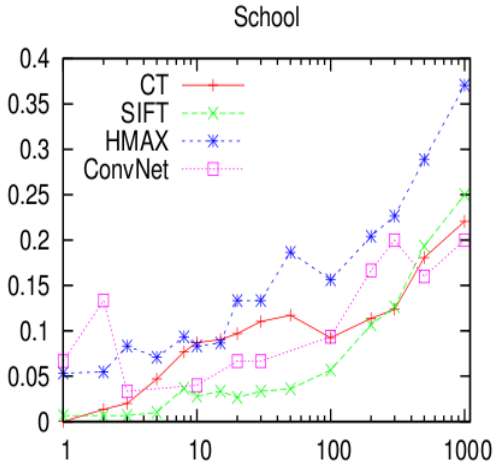


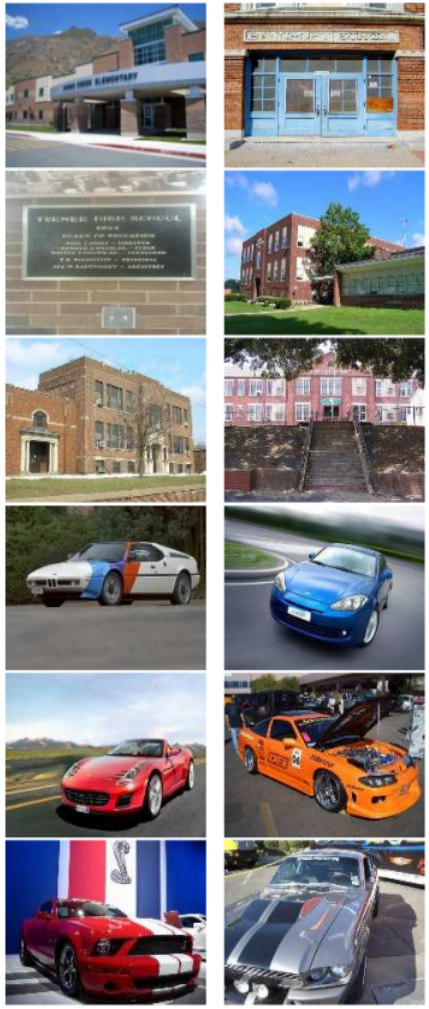


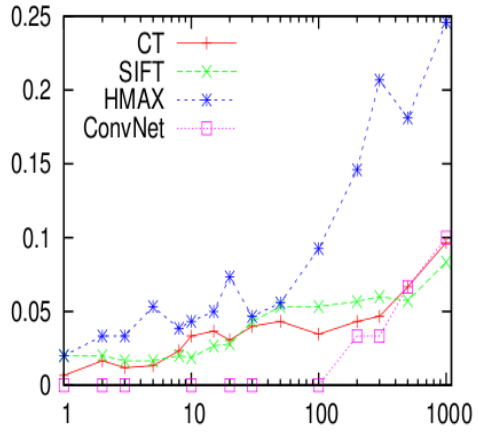


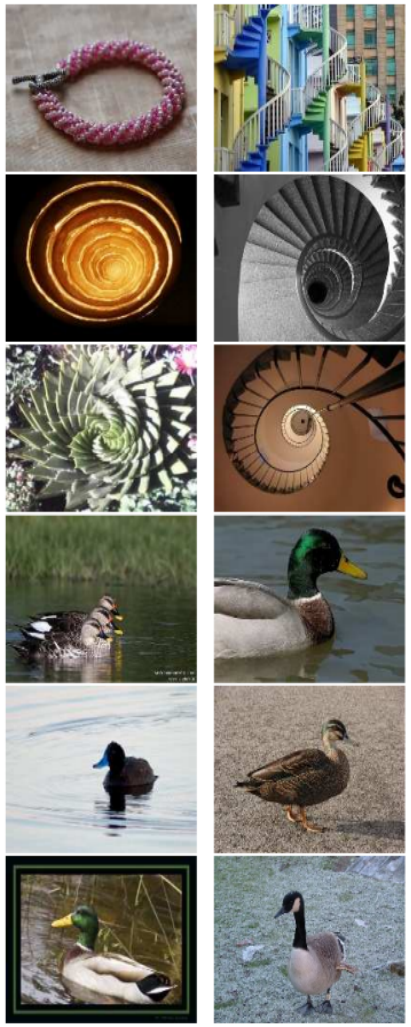


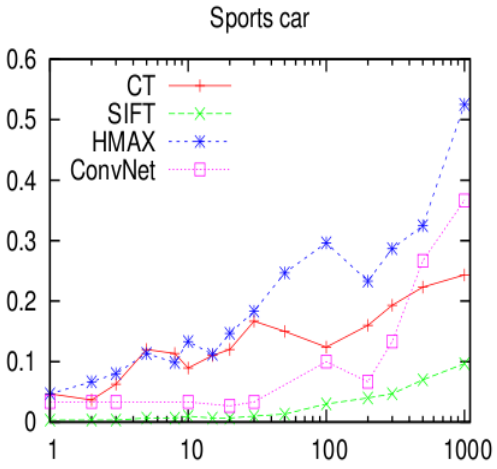
HMAX

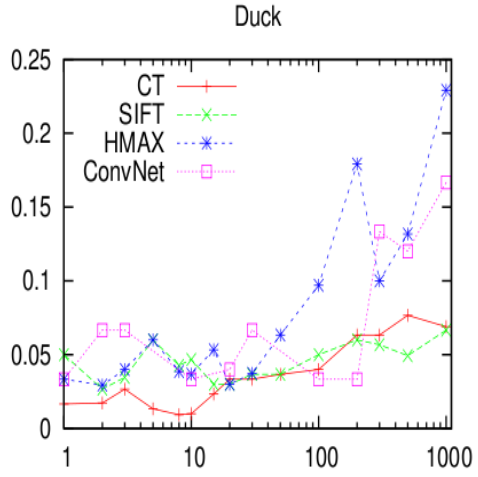

School
CT
SIFT
HMAX
ConvNet
0.4
0.35
0.3
0.25
0.2
0.15
0.1
0.05
0
1
10
100
1000



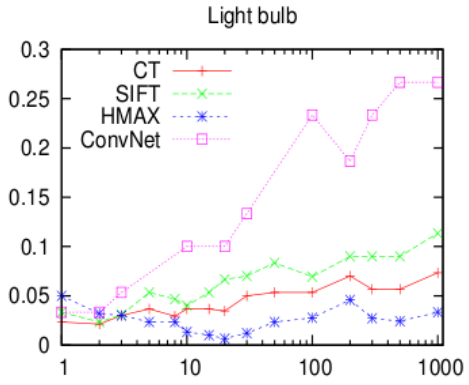

Coil
CT
SIFT
HMAX
ConvNet
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0.2
0.15
0.1
0.05
0
1
10
100
1000



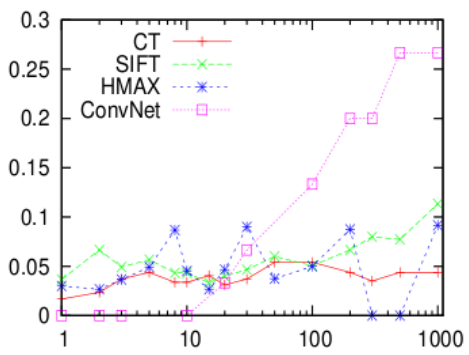

Sports car
CT
SIFT
HMAX
ConvNet
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0.5
0.4
0.3
0.2
0.1
0
1
10
100
1000

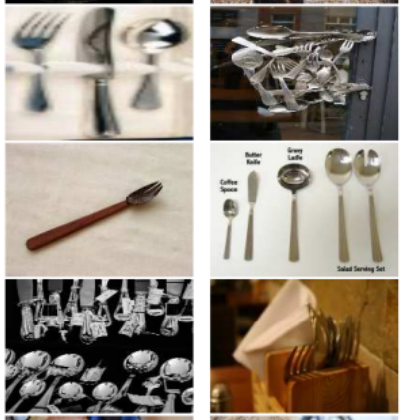

Duck
CT
SIFT
HMAX
ConvNet
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0.1
0.05
0
1
10
100
1000

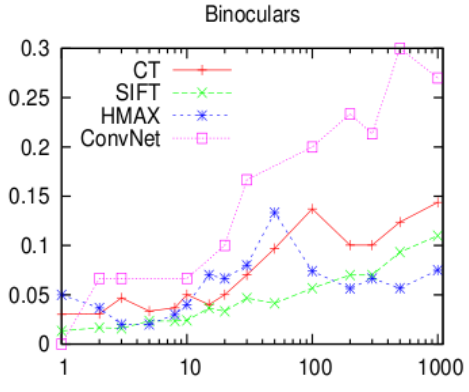
ConvNet: informative details



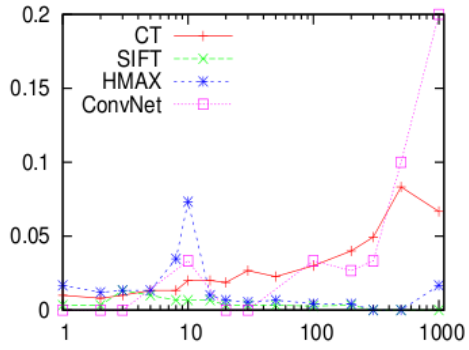




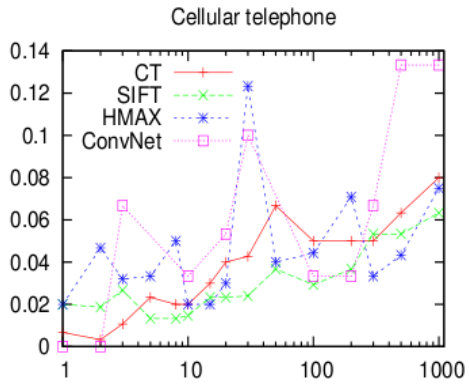




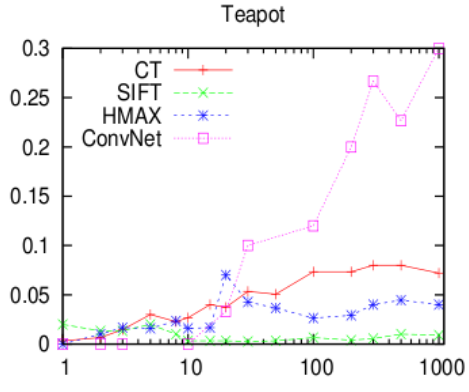






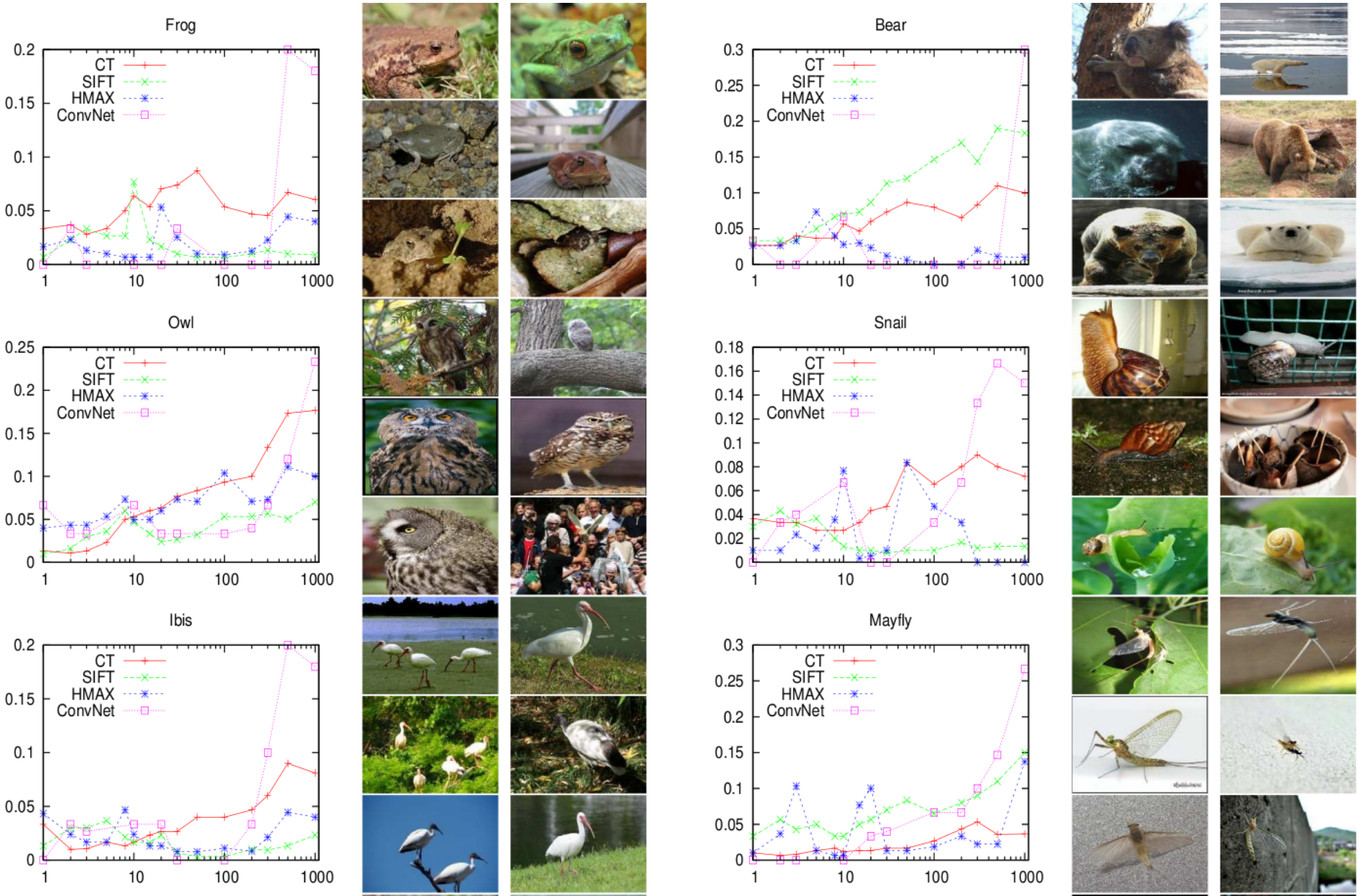








ConvNet: deformable objects



On Much Larger Dataset

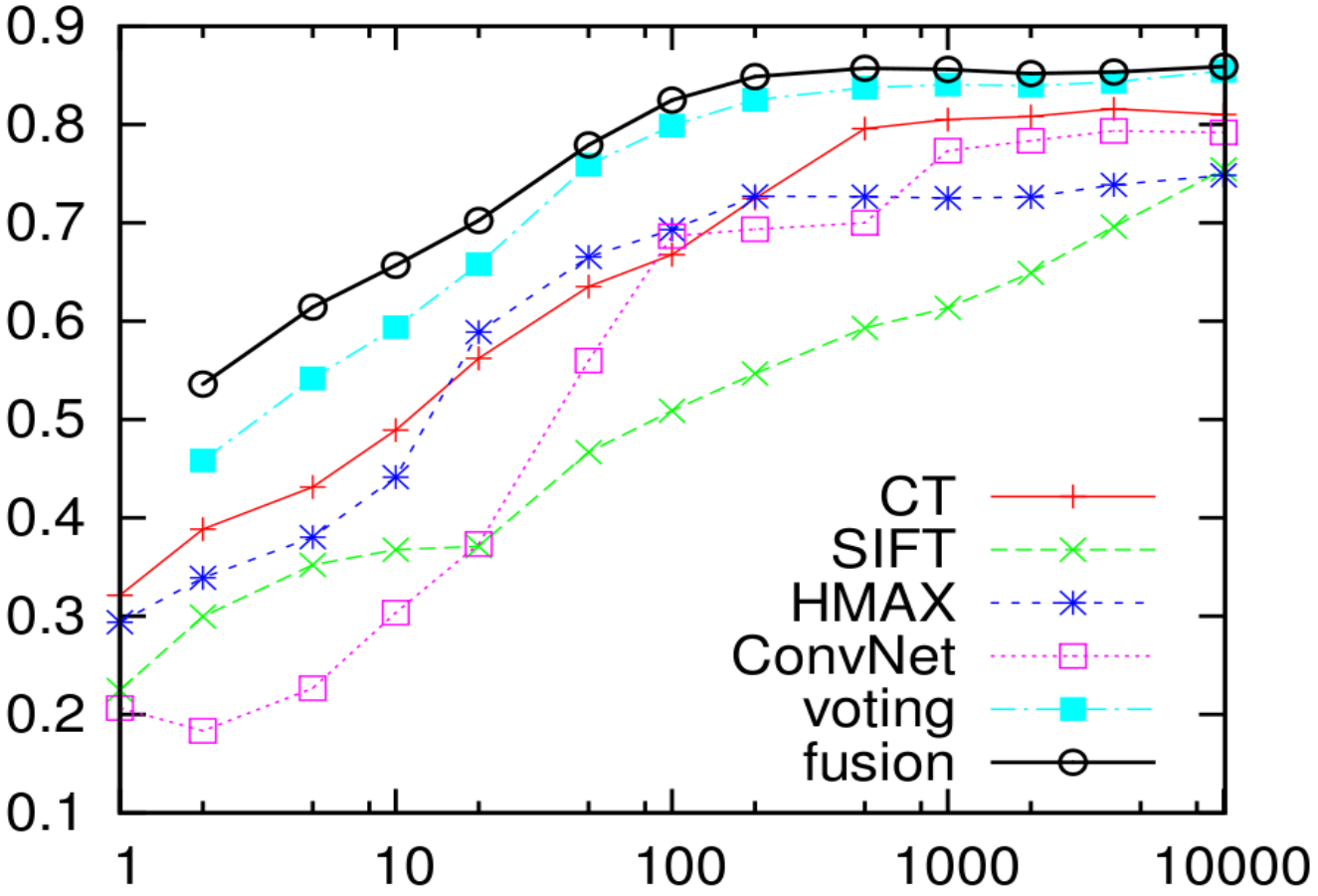
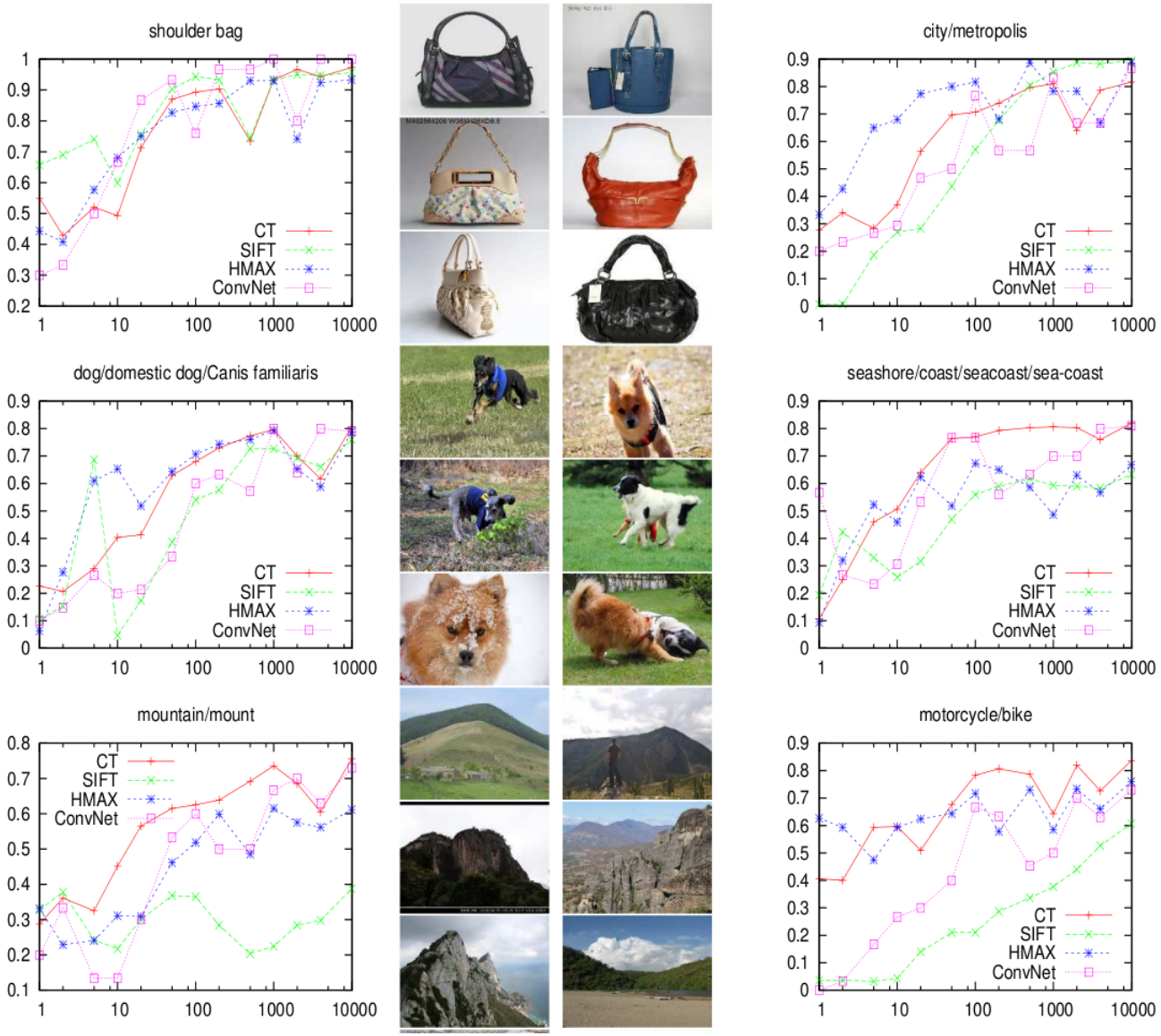
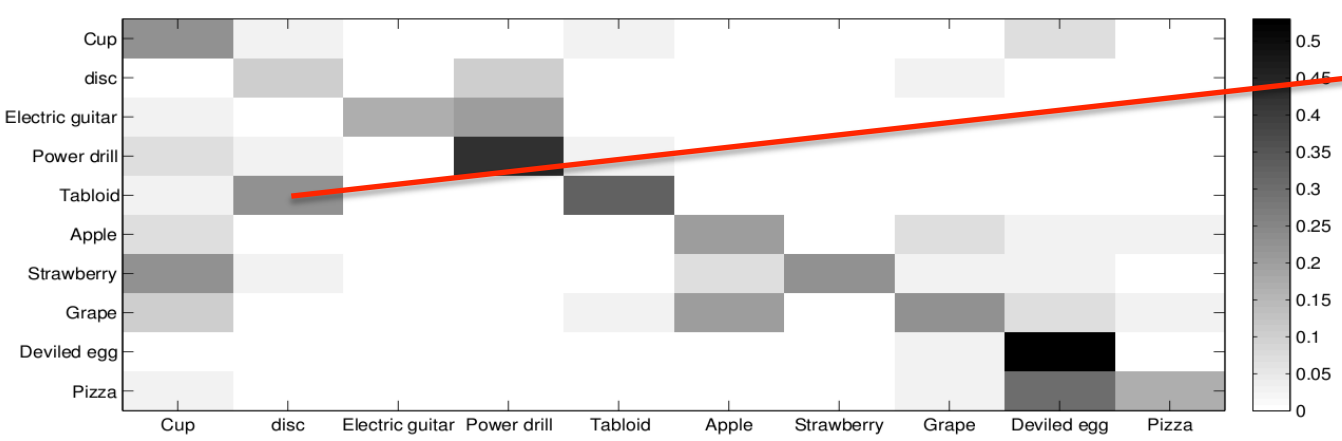


Fig. 14: Accuracy for PicasaWeb dataset. X-axis shows the number of training instances for each category. Y-axis shows the average accuracy.

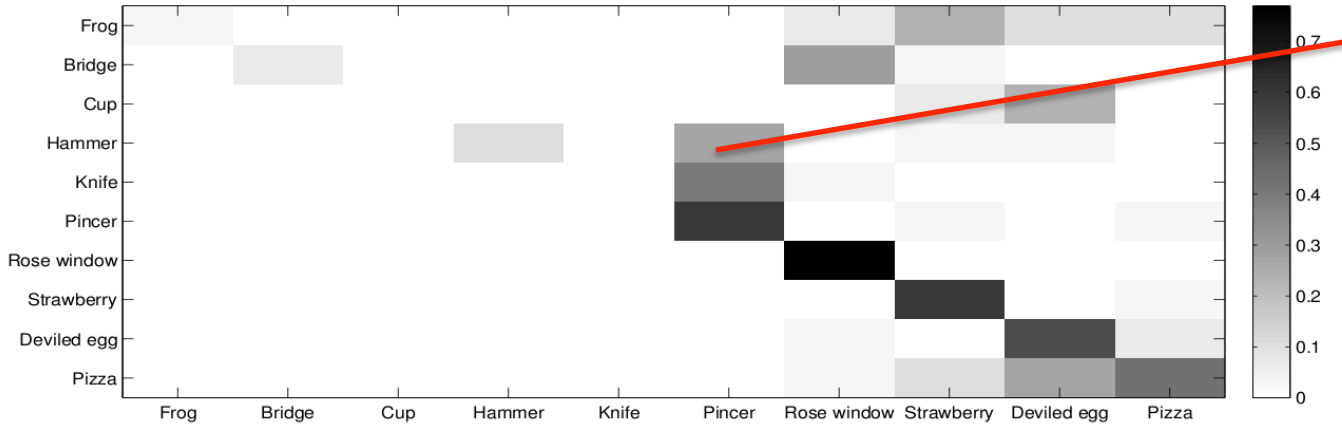
PicasaWeb: examples



Confusion matrix: CT & SIFT

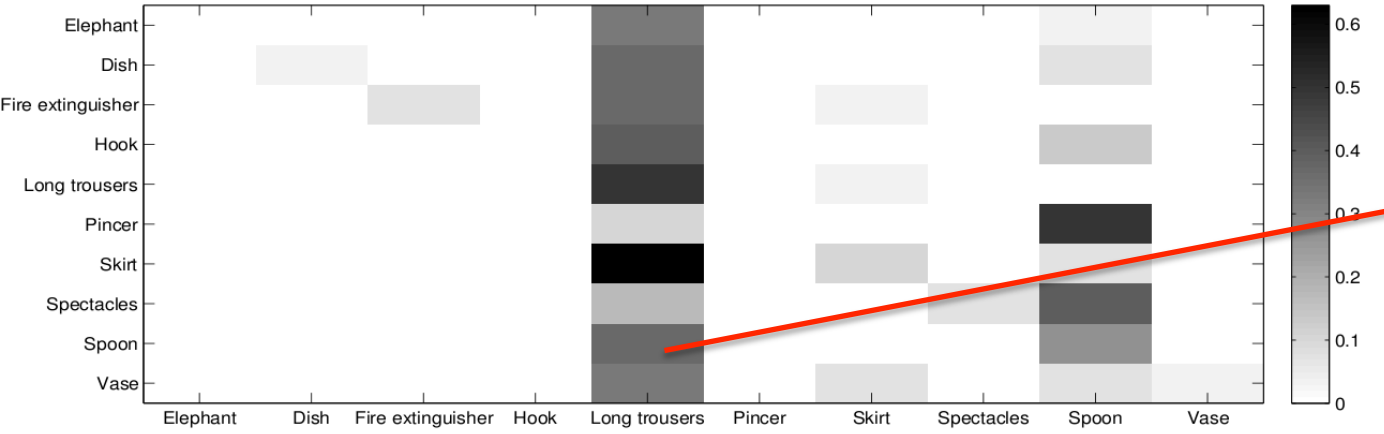








Confusion matrix: HMAX & ConvNet

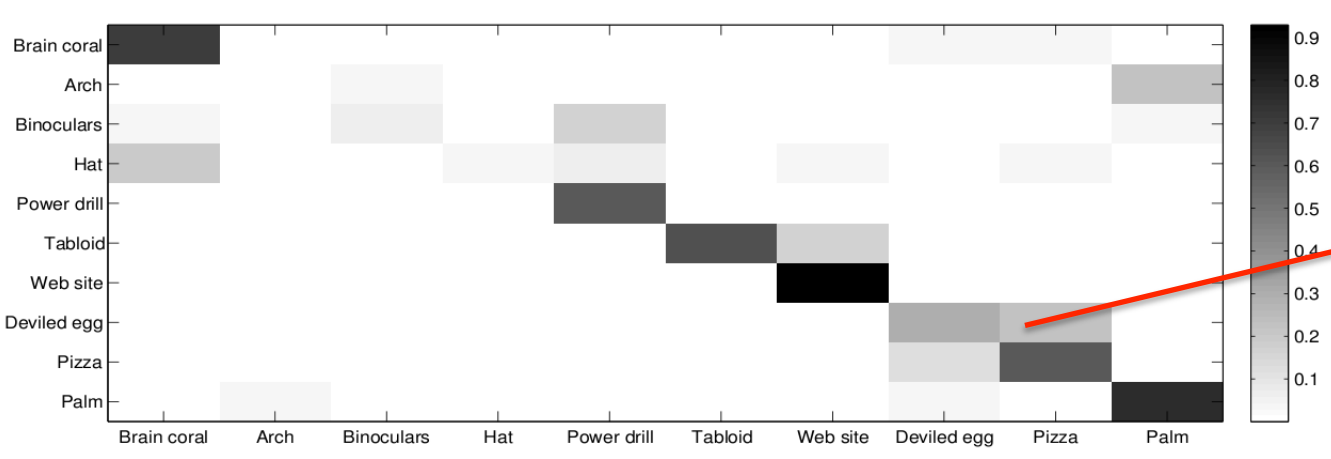








Confusion matrix: fusion





Deviled egg Pizza

More Data vs. Better Algorithms

Banko & Brill, 2001

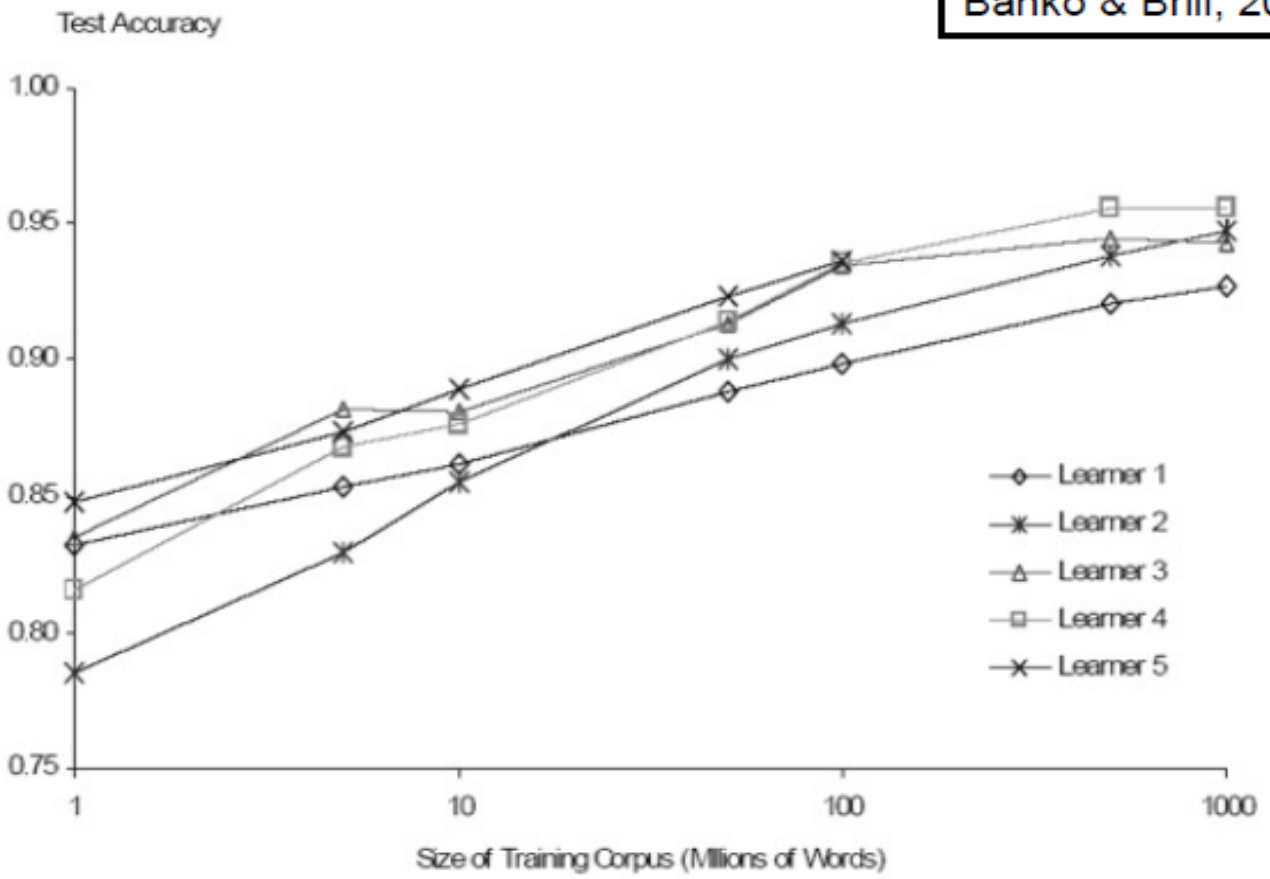


Figure 2. Learning Curves for Confusable Disambiguation

More Data vs. Better Algorithms

Banko & Brill, 2001

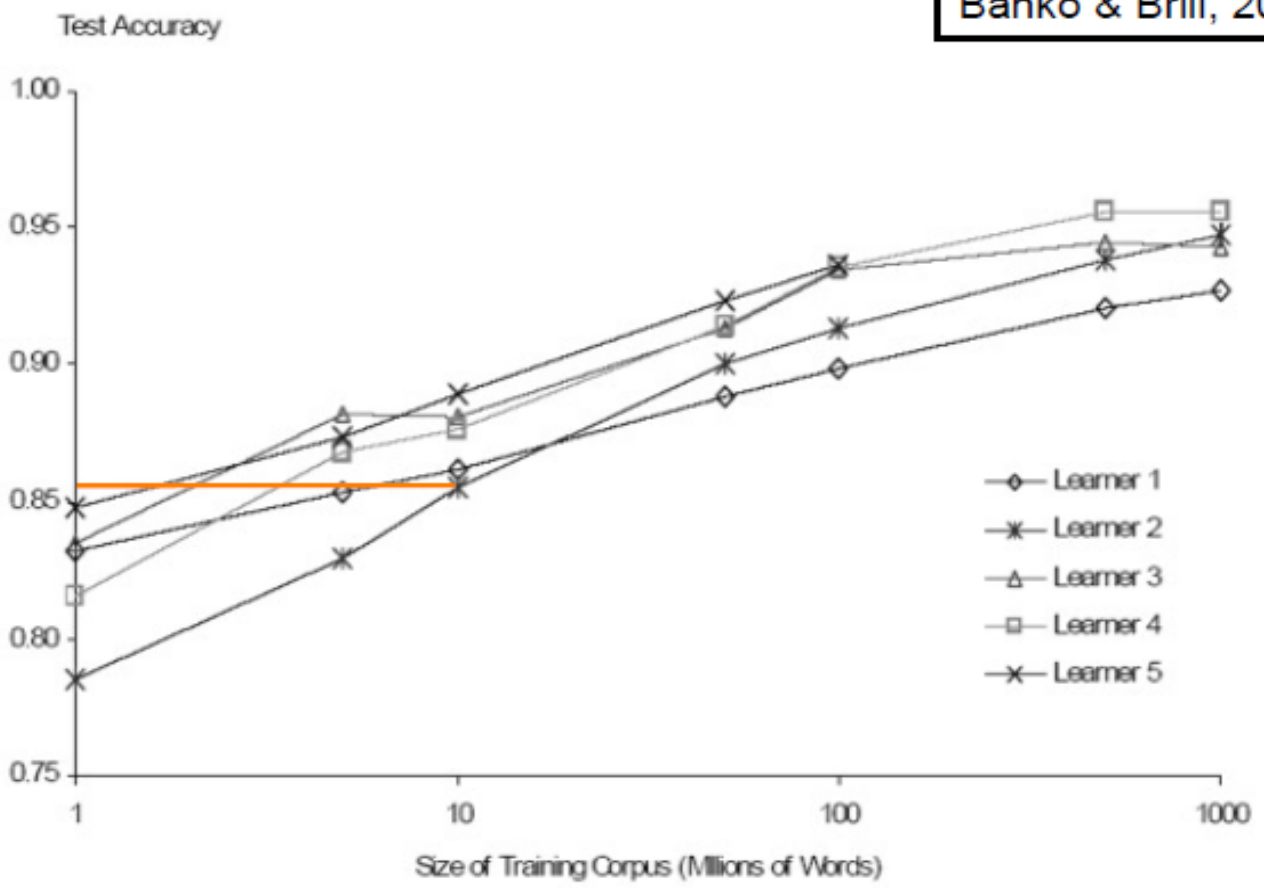


Figure 2. Learning Curves for Confusable Disambiguation

More Data vs. Better Algorithms

Banko & Brill, 2001

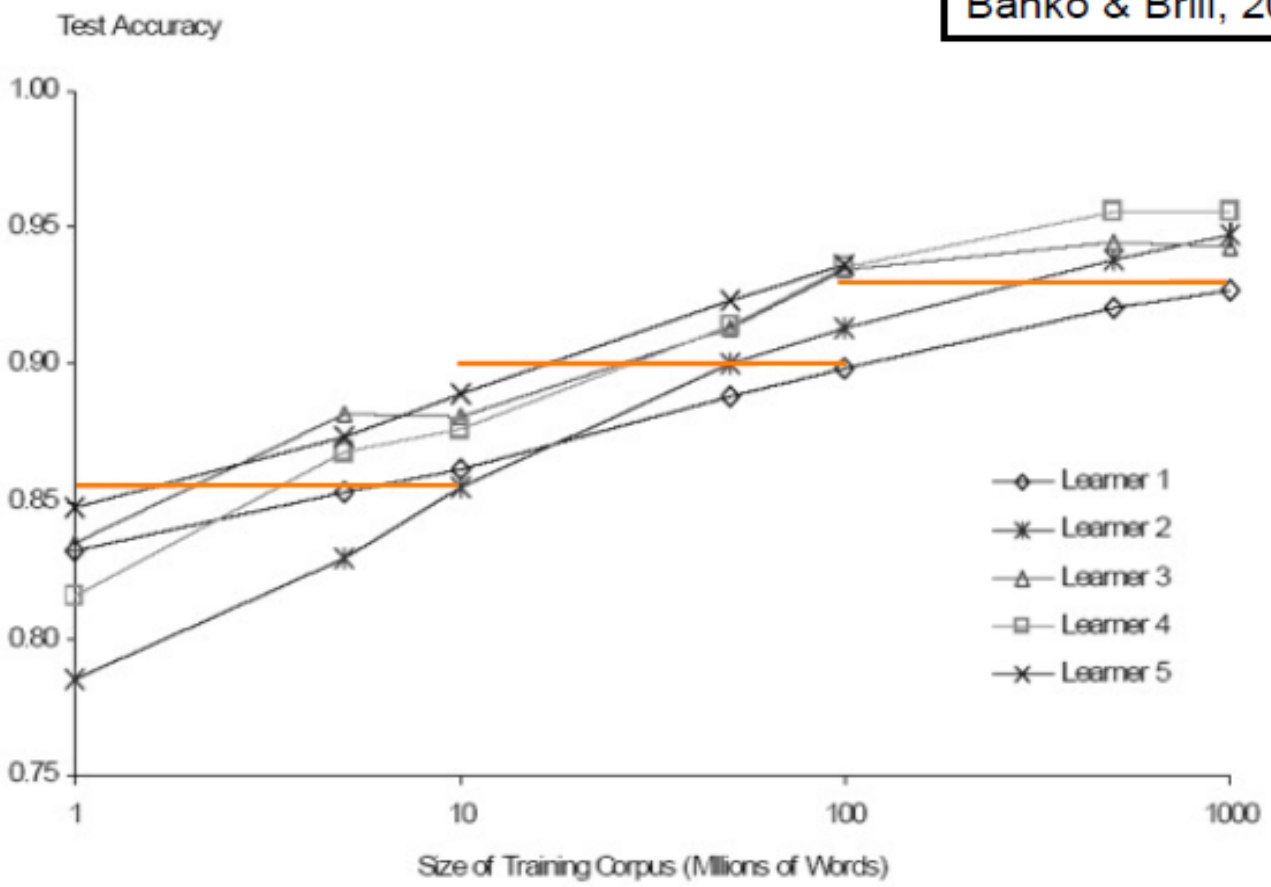
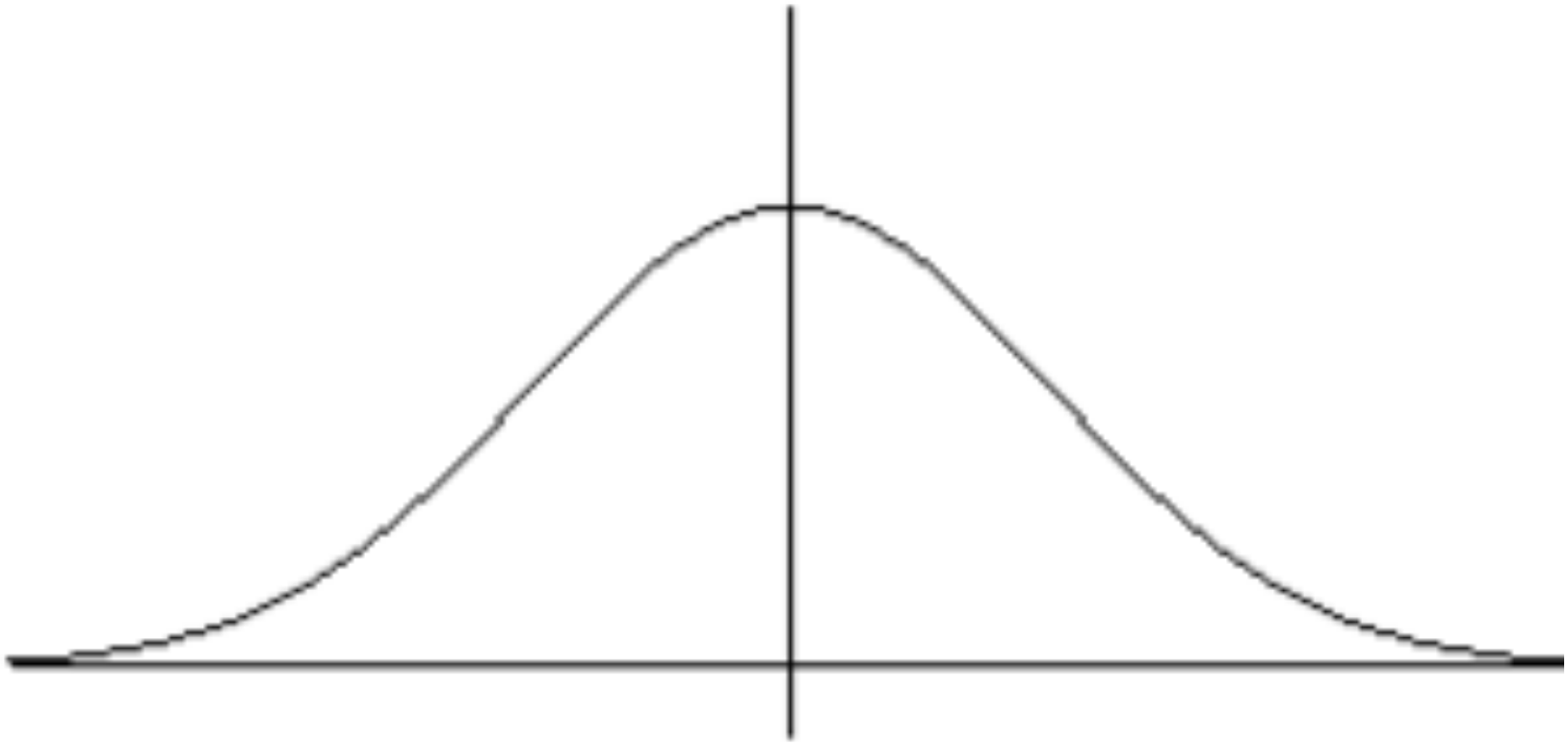


Figure 2. Learning Curves for Confusable Disambiguation

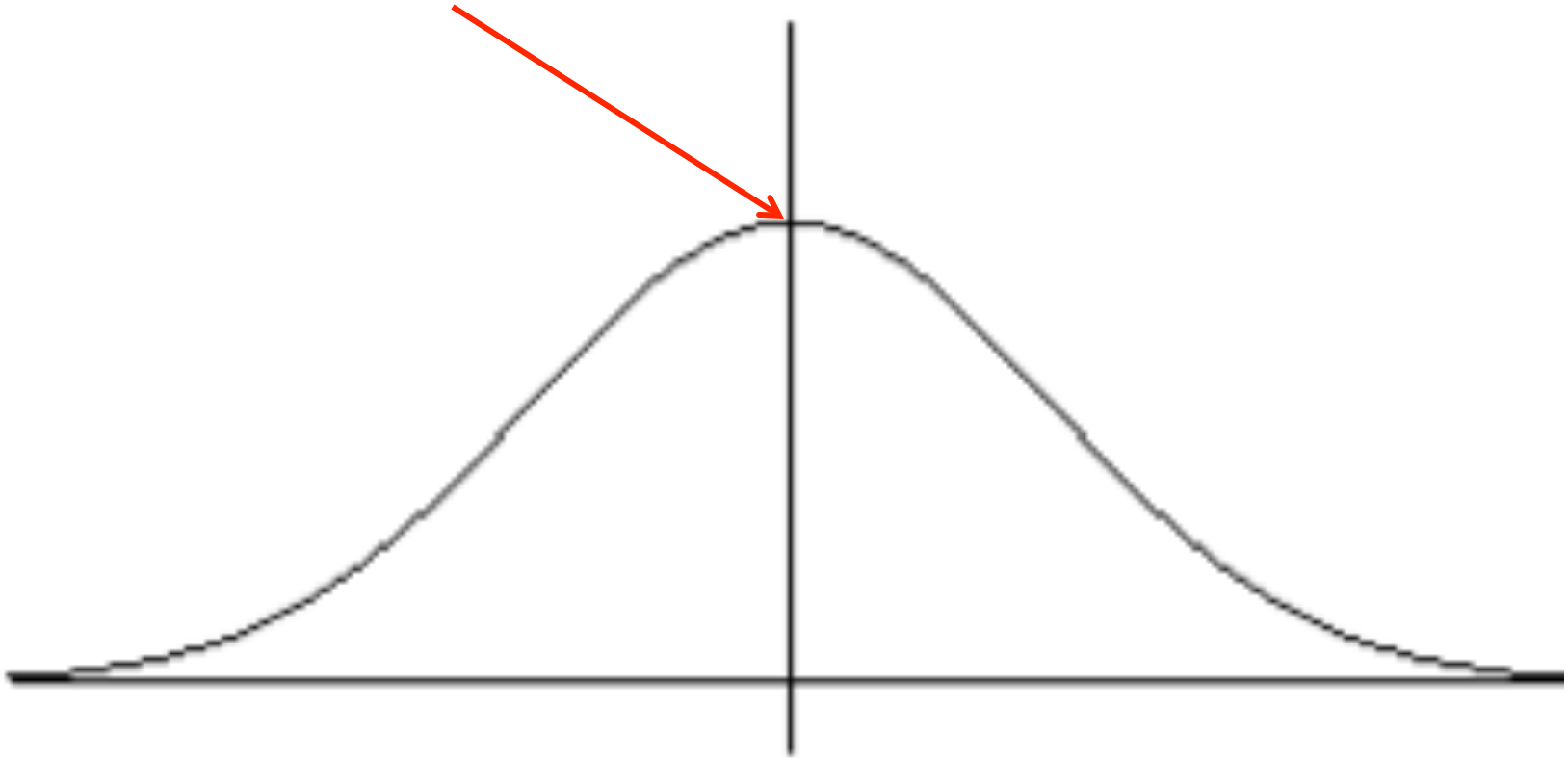
How Much Data?

Jerry Friedman



How Much Data?

Jerry Friedman



Feature Extraction Key Takeaways

- Balance tradeoffs between selectivity and invariance
- Bio-motivated methods seem to be productive
  - A semi-supervised approach
  - Clustering features in unsupervised way
  - Distill useful features in supervised way
- Models, simple or complex, requires more data to improve accuracy
- Simple model could do as well when data is abundant (and that's why large-scale learning)

More Data vs. Better Algorithms

Banko & Brill, 2001

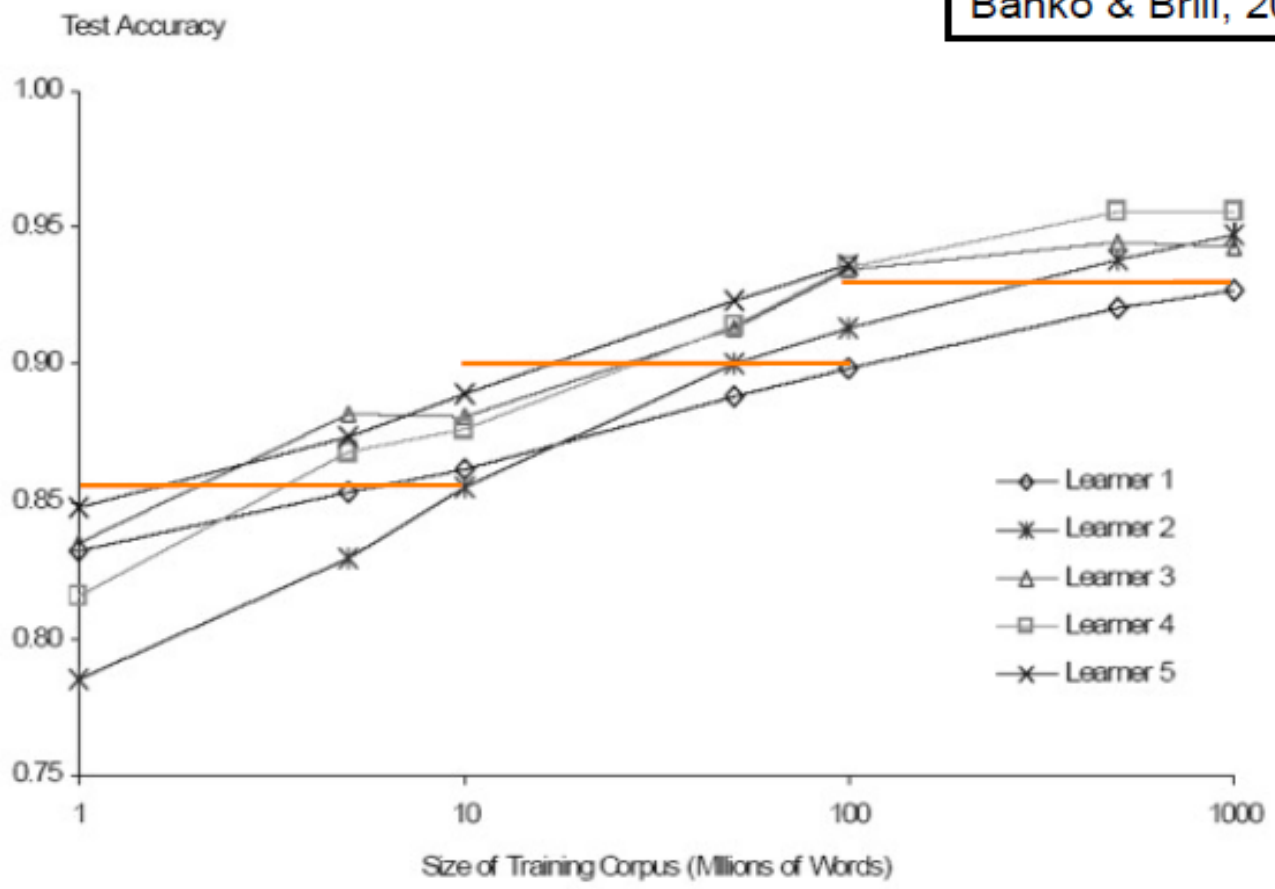


Figure 2. Learning Curves for Confusable Disambiguation

Reading

More references are cited by these

- ***A Data-Driven Study on Image Feature Extraction and Fusion***, Zhiyu Wang, Fangtao Li, Edward Y. Chang, and Shiqiang Yang, Google Technical Report, April 2012
- ***Foundations of Large-Scale Multimedia Information Management and Retrieval***, E. Y. Chang, Springer, 2011
- ***Convolutional deep belief networks for scalable unsupervised learning of hierarchical representations***, Honglak Lee, Roger Grosse, Rajesh Ranganath and Andrew Y. Ng. In *Proceedings of the Twenty-Sixth International Conference on Machine Learning,* 2009
- ***Robust Object Recognition with Cortex-like Mechanisms***, T. Serre, L. Wolf, S. Bileschi, M. Riesenhuber, and T. Poggio, IEEE Transactions on Pattern Analysis and Machine Intelligence, 29(3):411–426, 2007.
- ***Object Recognition from Local Scale-Invariant Features***, D.G. Lowe, In IEEE International Conference on Computer Vision (ICCV), 1999.